

# TM Forum Specification

## Geographic Site User Guide

**TMF674**

|                                                  |                                        |
|--------------------------------------------------|----------------------------------------|
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# Introduction

This document specifies the TMF674 REST API for Geographic Site Management. A **Geographic Site** is a specialization of the abstract **Place** entity, alongside **GeographicAddress** and **GeographicLocation**.

A Geographic Site represents a *managed* location, such as a retail store, warehouse or data centre. It will typically have associations with a collection of **relatedParty** that are in roles of responsibility for the Site. A large site might be described in terms of multiple relevant addresses and/or locations, both distinguished by **role** (such as a **delivery-address**, or **office-entrance**).

A Geographic Site can manage various unstructured site features, ranging from survey data to safety guidelines and hazard information.

This API is closely related to two other APIs that also cover the SID GeographicPlace entity:

- **Geographic Address** — manages places defined in a structured textual way (e.g., street name, street number) and validates the existence of a given address.
- **Geographic Location** — manages places defined as geometric objects in a spatial reference system (coordinates, boundaries).

## Supported Operations

The Geographic Site API supports the following operations:

- **Create** a new site
- **Partially update** an existing site
- **Retrieve** a list of sites, filtered by given criteria
- **Retrieve** a specific site
- **Notify** events related to a geographic site, including:
  - Site creation
  - Site removal
  - Site status change
  - Site value change (any data update within a site)

## Sample Use Cases

Below there are several use cases listed for the usage of the Geographic Site API.

### Field Service Management for Engineering Services

**Description:** A telecommunication company providing fixed fiber broadband services needs to efficiently manage field service operations for installation, maintenance, and repair of fiber optic infrastructure across various locations.

#### Maintenance and Routine Inspections:

**Description:** Regular maintenance and inspections are required to ensure the integrity and functionality of the fiber optic network. The system schedules routine maintenance inspections for fiber optic network sites.

Field service technicians receive notifications of scheduled maintenance tasks along with site details. Technicians use the Geographic Site API to retrieve site information. Technicians visit each site, perform maintenance tasks, conduct inspections, and update site status and maintenance records accordingly. Any issues or anomalies discovered during inspections are documented and reported for further investigation or repair.

### Hazard Identification and Mitigation

**Description:** Field service technicians need to be aware of potential hazards present at fiber optic network sites to ensure their safety and the integrity of the infrastructure.

Hazard information such as underground utilities, hazardous materials, or environmental risks is stored against geographic sites using the Geographic Site API. Before visiting a site for installation, maintenance, or repair tasks, field service technicians retrieve hazard information associated with the site. Technicians assess the potential risks and implement necessary safety measures and precautions to mitigate hazards during site visits.

By proactively addressing hazards, the company reduces the likelihood of accidents, injuries, and service disruptions, enhancing overall operational safety and efficiency.

### Emergency Response Planning

**Description:** In the event of emergencies such as natural disasters or infrastructure failures, field service teams need to quickly assess and respond to hazards affecting fiber optic network sites.

Hazard information including vulnerability to floods, earthquakes, or power outages is stored against geographic sites in the system. During emergency response planning, field

service managers use the GeographicSite API to identify high-risk sites and prioritize mitigation efforts. Field service technicians receive updated hazard information and emergency response protocols for affected sites.

In the event of an emergency, technicians can quickly assess hazards, prioritize safety measures, and coordinate with emergency responders to minimize service disruptions and ensure the safety of personnel and infrastructure.

## **Regulatory Compliance and Reporting**

**Description:** Telecommunication companies must comply with regulatory requirements related to workplace safety and environmental protection, including the management of hazardous materials and environmental risks.

Hazard information such as the presence of hazardous chemicals or protected habitats is stored against geographic sites in compliance with regulatory standards. During regulatory audits or inspections, field service managers use the GeographicSite API to retrieve comprehensive hazard data for reporting purposes.

Regulatory compliance reports are generated, documenting hazard identification, mitigation measures, and ongoing monitoring efforts at fiber optic network sites.

By maintaining accurate hazard records and demonstrating proactive risk management practices, the company ensures compliance with regulatory requirements and mitigates the risk of fines or penalties.

## **Franchise Management**

**Description:** A company with multiple franchise locations needs to efficiently manage and track all their sites.

The GeographicSite API enables the company to create, update, and retrieve information about each franchise location. They can easily add new franchise sites as the business expands, update site details such as operating hours or contact information, and retrieve lists of sites for reporting or analysis purposes.

Further examples of use cases are available in TM Forum Open Digital Architecture business scenarios and use cases (IG1228).

## Support of polymorphism and extension patterns

Support for polymorphic collections, types, and schema-based extensions is provided through a set of generic meta-attributes described below. Polymorphism in collections occurs when entities inherit from base entities. For instance, `GeographicSite`, `GeographicAddress`, and `GeographicLocation` inherit properties from the abstract `Place` entity.

Generic support for polymorphism and pattern extensions is detailed in the TMF API Guidelines, Part 2 (TMF630).

The `@type` attribute represents the actual class type of an entity. For example, within a list of `Place` instances, some may be instances of `GeographicSite` while others could be instances of `GeographicAddress` or `GeographicLocation`. The `@type` attribute provides this information. All resources and sub-resources of this API have `@type` attributes that can be provided when useful.

The `@referredType` attribute can be used within reference entities (such as a `GeographicAddress` object) to explicitly denote the actual entity type of the referred class. Note that in reference entities, the `@type` attribute, when used, denotes the class type of the reference itself, such as `AddressRef`, and not the class type of the referred object. However, since reference classes are rarely sub-classed, `@type` is generally not useful in reference objects.

The `@schemaLocation` property can be used in resources to specify user-defined properties of an entity or to specify the expected characteristics of an entity.

The `@baseType` attribute explicitly provides the base class of a given resource that has been extended.

# RESOURCE MODEL

## Managed Entity and Task Resource Models

### GeographicSite resource

A GeographicSite is a managed place, such as a retail store, warehouse or data centre. It will have related responsible parties, a calendar of access times and be described by addresses and/or locations.

### Resource model

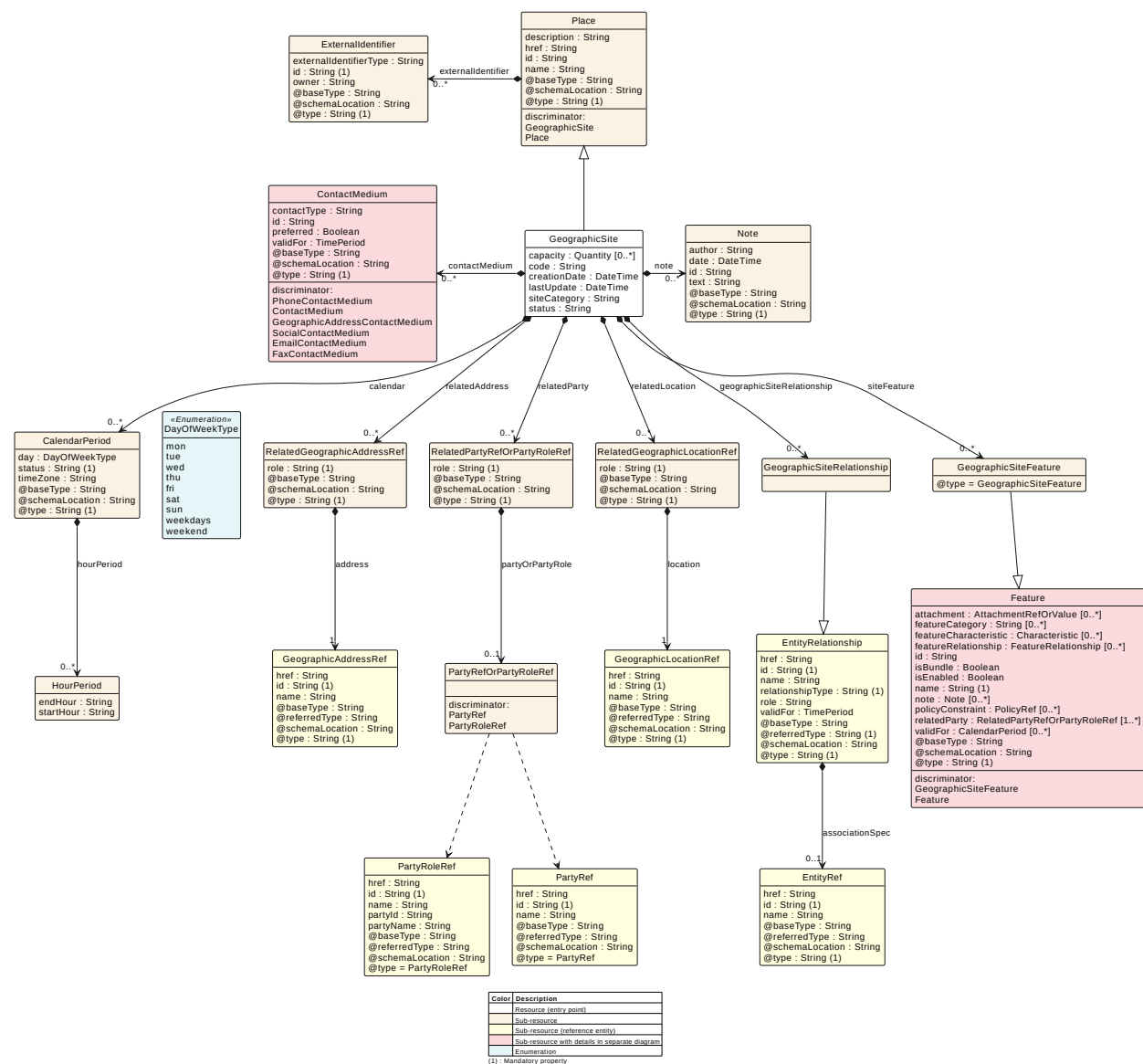


Figure 1 – GeographicSite

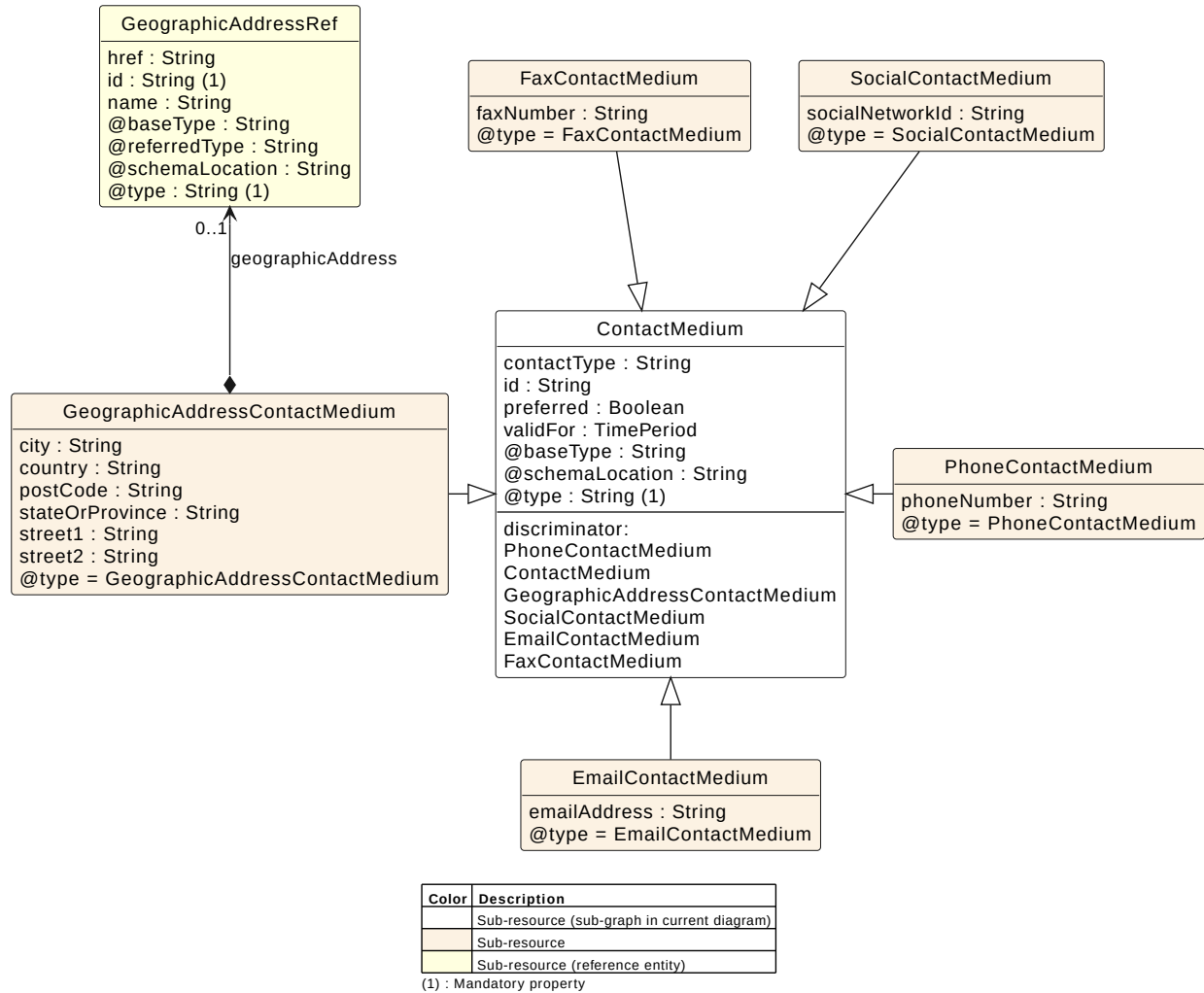


Figure 2 – ContactMedium

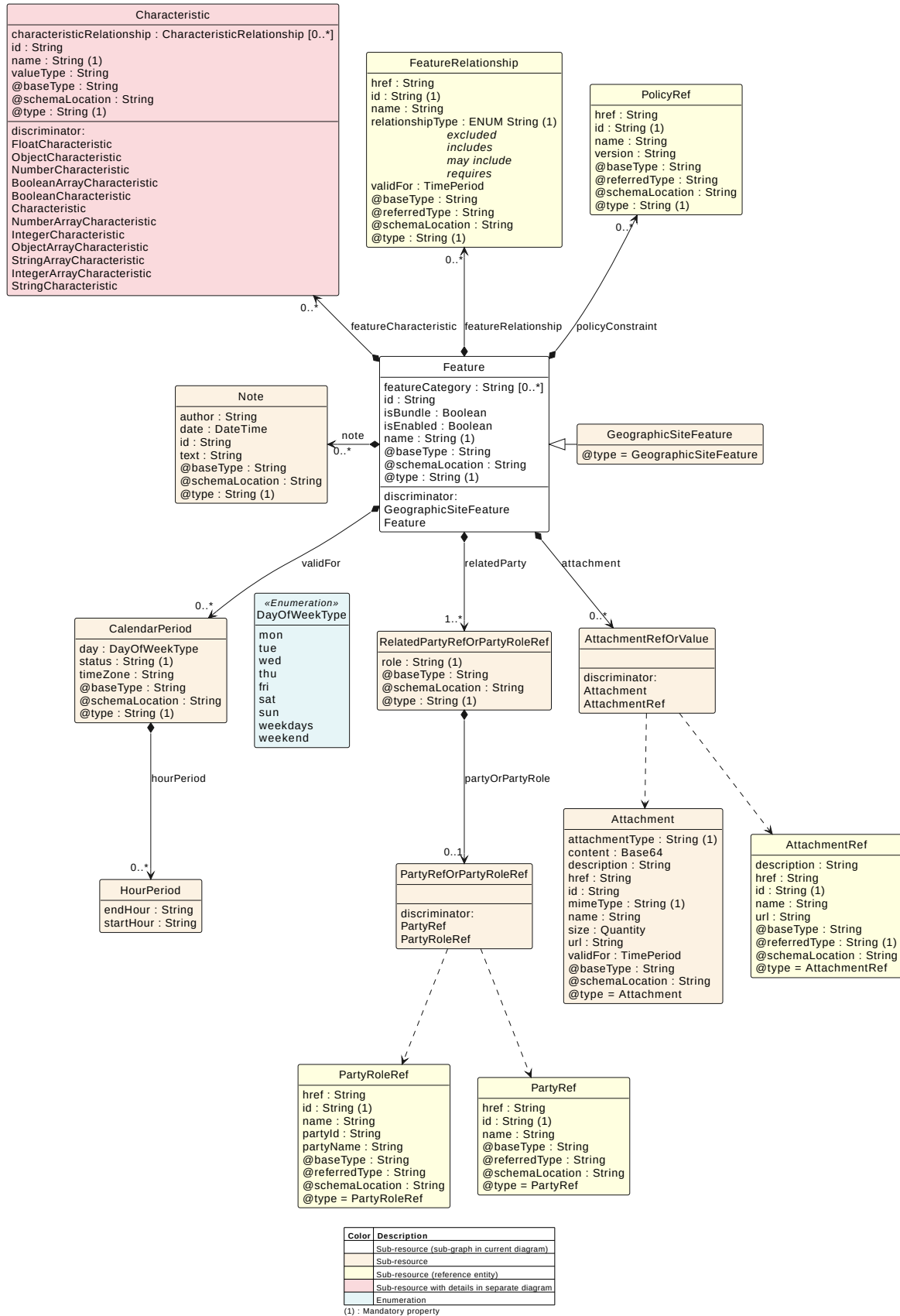


Figure 3 – Feature

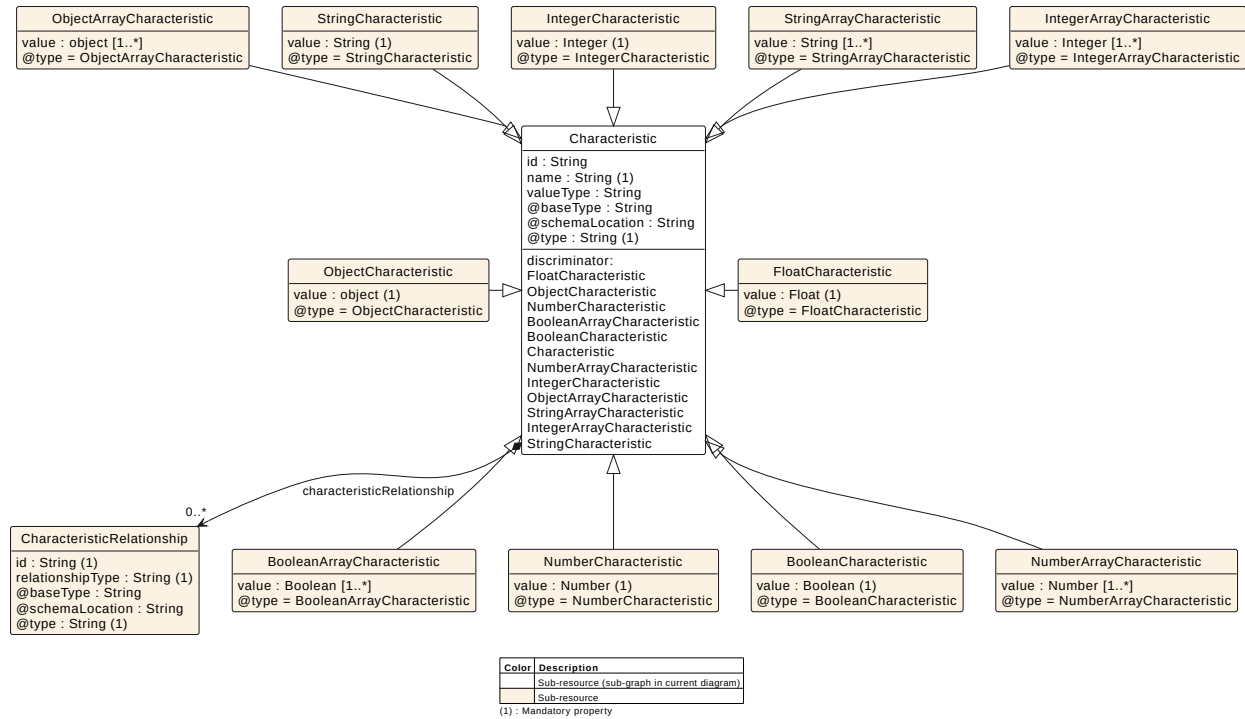


Figure 4 – Characteristic

## Field descriptions

### GeographicSite fields

|          |                                                                                                                                                                                                          |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| calendar | A CalendarPeriod [0..*]. Periods of (for example) access, opening, alarm, unavailability of this site.                                                                                                   |
| capacity | A Quantity [0..*]. Capacities for the site indicating maximum operational thresholds (for example: people, floorspace, racks).                                                                           |
| code     | <p>DEPRECATED</p> <p>A String. Deprecated: Use id, externalIdentifier and/or siteCategory or name.</p> <div style="border: 1px solid black; padding: 5px; margin-top: 10px;"> <p>example: BTS</p> </div> |

|                            |                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| contactMedium              | <p>A ContactMedium [0..*]. Contact information for people with responsibility for this site.</p> <p>ContactMedium can be instantiated as</p> <ul style="list-style-type: none"> <li>* EmailContactMedium</li> <li>* FaxContactMedium</li> <li>* GeographicAddressContactMedium</li> <li>* PhoneContactMedium</li> <li>* SocialContactMedium</li> </ul> |
| creationDate               | <p>A DateTime. Date and time when the GeographicSite was created.</p> <p>example: 2024-09-23T00:00:00Z</p>                                                                                                                                                                                                                                             |
| description                | <p>A String. Additional information of a user-friendly description of the Place.</p> <p>example: A three-storey office building</p>                                                                                                                                                                                                                    |
| externalIdentifier         | <p>An ExternalIdentifier [0..*]. Collection of external identifiers: This Place may be known to external systems by other identifiers.</p>                                                                                                                                                                                                             |
| geographicSiteRelationship | <p>A GeographicSiteRelationship [0..*]. This site might have relationships to other sites.</p>                                                                                                                                                                                                                                                         |
| href                       | <p>A String. Hyperlink reference.</p>                                                                                                                                                                                                                                                                                                                  |
| id                         | <p>A String. Unique identifier.</p>                                                                                                                                                                                                                                                                                                                    |
| lastUpdate                 | <p>A DateTime. Date and time when the GeographicSite was last updated.</p> <p>example: 2024-10-03T00:00:00Z</p>                                                                                                                                                                                                                                        |
| name                       | <p>A String. A user-friendly name for the Place.</p> <p>example: Fountain House</p>                                                                                                                                                                                                                                                                    |
| note                       | <p>A Note [0..*]. Unstructured notes made about this site.</p>                                                                                                                                                                                                                                                                                         |

|                 |                                                                                                                                                 |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| relatedAddress  | A RelatedGeographicAddressRef [0..*]. Collection of related address references (such as delivery-address, office-address).                      |
| relatedLocation | A RelatedGeographicLocationRef [0..*]. Collection of related location references (such as retail-complex, service-access).                      |
| relatedParty    | A RelatedPartyRefOrPartyRoleRef [0..*]. Collection of related roles and parties references (such as site manager, site owner, site operator).   |
| siteCategory    | A String. Site classification/category.<br><br>example: ShoppingUnit                                                                            |
| siteFeature     | A GeographicSiteFeature [0..*]. Collection of site features.                                                                                    |
| status          | A String. The condition of the GeographicSite, such as planned, underConstruction, cancelled, active, inactive, former.<br><br>example: planned |
| @baseType       | A String. When sub-classing, this defines the super-class.                                                                                      |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships.                                                     |
| @type           | A String [1]. When sub-classing, this defines the sub-class Extensible name.                                                                    |

### CalendarPeriod sub-resource fields

The CalendarPeriod schema represents a period of time within a calendar, defining various attributes such as the applicable day, timezone, hour period, and status. It is designed to capture information about the availability or status of a calendar period, which can be used in scheduling, booking, or resource allocation applications.

|                 |                                                                                                                                                                              |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| day             | A DayOfWeekType. Possible values for the days of the week. ENUMERATED with values:<br>* mon<br>* tue<br>* wed<br>* thu<br>* fri<br>* sat<br>* sun<br>* weekdays<br>* weekend |
| hourPeriod      | A HourPeriod [0..*]. Start and end time range in 24-hour HH:mm format.                                                                                                       |
| status          | A String [1]. Indication of the availability of the calendar period (e.g.: available, booked, etc.).                                                                         |
| timeZone        | A String. Indication of the timezone applicable to the calendar information in ISO 8601 timezone: either 'Z' for UTC or an offset like +02:00 or -05:30.                     |
| @baseType       | A String. When sub-classing, this defines the super-class.                                                                                                                   |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships.                                                                                  |
| @type           | A String [1]. When sub-classing, this defines the sub-class Extensible name.                                                                                                 |

### HourPeriod sub-resource fields

Hour interval.

|           |                                                                                                                                       |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------|
| endHour   | A String. The end of the hour period, expressed in 24-hour HH:mm format (e.g., 12:00 for noon, 23:59 for one minute before midnight). |
| startHour | A String. The beginning of the hour period, expressed in 24-hour HH:mm format (e.g., 09:00 for 9 AM, 17:30 for 5:30 PM).              |

### ContactMedium sub-resource fields

Indicates the contact medium that could be used to contact the party. This is an abstract base class, the actual value is in one of the strongly-typed subclasses :

EmailContactMedium, FaxContactMedium, PhoneContactMedium,  
GeographicAddressContactMedium, SocialMediaContactMedium...

|                   |                                                                                                                                                                         |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| contactType       | A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.                                 |
| id                | A String. Identifier for this contact medium.                                                                                                                           |
| preferred         | A Boolean. If true, indicates that is the preferred contact medium.                                                                                                     |
| validFor          | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.                                                                  |
| @baseType         | A String. When sub-classing, this defines the super-class.                                                                                                              |
| @schemaLocation   | A String. A URI to a JSON-Schema file that defines additional attributes and relationships.                                                                             |
| @type             | A String [1]. When sub-classing, this defines the sub-class Extensible name.                                                                                            |
|                   | ContactMedium can be instantiated as<br>* EmailContactMedium<br>* FaxContactMedium<br>* GeographicAddressContactMedium<br>* PhoneContactMedium<br>* SocialContactMedium |
| city              | This property is present in subclasses                                                                                                                                  |
| country           | This property is present in subclasses                                                                                                                                  |
| emailAddress      | This property is present in subclasses                                                                                                                                  |
| faxNumber         | This property is present in subclasses                                                                                                                                  |
| geographicAddress | This property is present in subclasses                                                                                                                                  |
| phoneNumber       | This property is present in subclasses                                                                                                                                  |
| postCode          | This property is present in subclasses                                                                                                                                  |
| socialNetworkId   | This property is present in subclasses                                                                                                                                  |
| stateOrProvince   | This property is present in subclasses                                                                                                                                  |
| street1           | This property is present in subclasses                                                                                                                                  |
| street2           | This property is present in subclasses                                                                                                                                  |

### GeographicAddressRef sub-resource fields

Reference to a Geographic Address.

|                 |                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------|
| href            | A String. Hyperlink reference.                                                              |
| id              | A String [1]. Unique identifier.                                                            |
| name            | A String. Name of the referred entity.                                                      |
| @baseType       | A String. When sub-classing, this defines the super-class.                                  |
| @referredType   | A String. The actual type of the target instance when needed for disambiguation.            |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type           | A String [1]. When sub-classing, this defines the sub-class Extensible name.                |

#### EmailContactMedium sub-resource fields

Describes an email that could be used to contact a party (an individual or an organization).

|                 |                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| contactType     | A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used. |
| emailAddress    | A String. Full email address in standard format.                                                                                        |
| id              | A String. Identifier for this contact medium.                                                                                           |
| preferred       | A Boolean. If true, indicates that is the preferred contact medium.                                                                     |
| validFor        | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.                                  |
| @baseType       | A String. When sub-classing, this defines the super-class.                                                                              |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships.                                             |
| @type           | A String [1]. When sub-classing, this defines the sub-class Extensible name.                                                            |

#### FaxContactMedium sub-resource fields

Describes a fax that could be used to contact a party (an individual or an organization).

|                 |                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| contactType     | A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used. |
| faxNumber       | A String. The fax number of the contact.                                                                                                |
| id              | A String. Identifier for this contact medium.                                                                                           |
| preferred       | A Boolean. If true, indicates that is the preferred contact medium.                                                                     |
| validFor        | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.                                  |
| @baseType       | A String. When sub-classing, this defines the super-class.                                                                              |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships.                                             |
| @type           | A String [1]. When sub-classing, this defines the sub-class Extensible name.                                                            |

### GeographicAddressContactMedium sub-resource fields

Describes a geographical address that could be used to contact a party (an individual or an organization).

|                   |                                                                                                                                         |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| city              | A String. The city.                                                                                                                     |
| contactType       | A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used. |
| country           | A String. The country.                                                                                                                  |
| geographicAddress | A GeographicAddressRef. Reference to a Geographic Address.                                                                              |
| id                | A String. Identifier for this contact medium.                                                                                           |
| postCode          | A String. Postcode.                                                                                                                     |
| preferred         | A Boolean. If true, indicates that is the preferred contact medium.                                                                     |
| stateOrProvince   | A String. State or province.                                                                                                            |
| street1           | A String. Describes the street.                                                                                                         |
| street2           | A String. Complementary street description.                                                                                             |

|                 |                                                                                                        |
|-----------------|--------------------------------------------------------------------------------------------------------|
| validFor        | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType       | A String. When sub-classing, this defines the super-class.                                             |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships.            |
| @type           | A String [1]. When sub-classing, this defines the sub-class Extensible name.                           |

### PhoneContactMedium sub-resource fields

Describes a phone number that could be used to contact a party (an individual or an organization).

|                 |                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| contactType     | A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used. |
| id              | A String. Identifier for this contact medium.                                                                                           |
| phoneNumber     | A String. The phone number of the contact.                                                                                              |
| preferred       | A Boolean. If true, indicates that is the preferred contact medium.                                                                     |
| validFor        | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.                                  |
| @baseType       | A String. When sub-classing, this defines the super-class.                                                                              |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships.                                             |
| @type           | A String [1]. When sub-classing, this defines the sub-class Extensible name.                                                            |

### SocialContactMedium sub-resource fields

Describes a social media identifier that could be used to contact a party (an individual or an organization).

|             |                                                                                                                                         |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| contactType | A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used. |
| id          | A String. Identifier for this contact medium.                                                                                           |

|                 |                                                                                                        |
|-----------------|--------------------------------------------------------------------------------------------------------|
| preferred       | A Boolean. If true, indicates that is the preferred contact medium.                                    |
| socialNetworkId | A String. Identifier as a member of a social network.                                                  |
| validFor        | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType       | A String. When sub-classing, this defines the super-class.                                             |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships.            |
| @type           | A String [1]. When sub-classing, this defines the sub-class Extensible name.                           |

### ExternalIdentifier sub-resource fields

An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.

|                        |                                                                                                             |
|------------------------|-------------------------------------------------------------------------------------------------------------|
| externalIdentifierType | A String. Type of the identification, typically would be the type of the entity within the external system. |
| id                     | A String [1]. Identification of the entity within the external system.                                      |
| owner                  | A String. Name of the external system that owns the entity.                                                 |
| @baseType              | A String. When sub-classing, this defines the super-class.                                                  |
| @schemaLocation        | A String. A URI to a JSON-Schema file that defines additional attributes and relationships.                 |
| @type                  | A String [1]. When sub-classing, this defines the sub-class Extensible name.                                |

### GeographicSiteRelationship sub-resource fields

Details of a geographic site relationship.

|                  |                                                                                                        |
|------------------|--------------------------------------------------------------------------------------------------------|
| associationSpec  | An EntityRef. Entity reference schema to be use for all entityRef class.                               |
| href             | A String.                                                                                              |
| id               | A String [1].                                                                                          |
| name             | A String.                                                                                              |
| relationshipType | A String [1]. Type of relationship between GeographicSites, such as adjacentTo, connectedBy.           |
| role             | A String. The association role for this entity.                                                        |
| validFor         | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType        | A String. When sub-classing, this defines the super-class.                                             |
| @referredType    | A String [1].                                                                                          |
| @schemaLocation  | A String. A URI to a JSON-Schema file that defines additional attributes and relationships.            |
| @type            | A String [1].                                                                                          |

### EntityRef sub-resource fields

Entity reference schema to be use for all entityRef class.

|                 |                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------|
| href            | A String. The URI of the referred entity.                                                   |
| id              | A String [1]. The identifier of the referred entity.                                        |
| name            | A String. Name of the referred entity.                                                      |
| @baseType       | A String. When sub-classing, this defines the super-class.                                  |
| @referredType   | A String. The actual type of the target instance when needed for disambiguation.            |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type           | A String [1]. When sub-classing, this defines the sub-class Extensible name.                |

### EntityRelationship sub-resource fields

A uni-directional relationship from this entity to a target entity instance.

|                  |                                                                                                        |
|------------------|--------------------------------------------------------------------------------------------------------|
| associationSpec  | An EntityRef. Entity reference schema to be use for all entityRef class.                               |
| href             | A String.                                                                                              |
| id               | A String [1].                                                                                          |
| name             | A String.                                                                                              |
| relationshipType | A String [1]. Type of relationship such as migration, substitution, dependency, exclusivity.           |
| role             | A String. The association role for this entity.                                                        |
| validFor         | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType        | A String. When sub-classing, this defines the super-class.                                             |
| @referredType    | A String [1].                                                                                          |
| @schemaLocation  | A String. A URI to a JSON-Schema file that defines additional attributes and relationships.            |
| @type            | A String [1].                                                                                          |
|                  | EntityRelationship can be instantiated as<br>* GeographicSiteRelationship                              |

### Note sub-resource fields

Extra information about a given entity.

|                 |                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------|
| author          | A String. Author of the note.                                                               |
| date            | A DateTime. Date of the note.                                                               |
| id              | A String. Identifier of the note within its containing entity.                              |
| text            | A String. Text of the note.                                                                 |
| @baseType       | A String. When sub-classing, this defines the super-class.                                  |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type           | A String [1]. When sub-classing, this defines the sub-class<br>Extensible name.             |

### RelatedGeographicAddressRef sub-resource fields

A Postal Address (by reference) with its associated role.

|                 |                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------|
| address         | A GeographicAddressRef [1]. Reference to a Geographic Address.                              |
| role            | A String [1]. The context in which this Address is being used.                              |
| @baseType       | A String. When sub-classing, this defines the super-class.                                  |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type           | A String [1]. When sub-classing, this defines the sub-class Extensible name.                |

### RelatedGeographicLocationRef sub-resource fields

A Geographic Location (by reference) with its associated role.

|                 |                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------|
| location        | A GeographicLocationRef [1]. Reference to a Geographic Location.                            |
| role            | A String [1]. The context in which this location is being used.                             |
| @baseType       | A String. When sub-classing, this defines the super-class.                                  |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type           | A String [1]. When sub-classing, this defines the sub-class Extensible name.                |

### GeographicLocationRef sub-resource fields

Reference to a Geographic Location.

|                 |                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------|
| href            | A String. Hyperlink reference.                                                              |
| id              | A String [1]. Unique identifier.                                                            |
| name            | A String. Name of the referred entity.                                                      |
| @baseType       | A String. When sub-classing, this defines the super-class.                                  |
| @referredType   | A String. The actual type of the target instance when needed for disambiguation.            |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |

|       |                                                                              |
|-------|------------------------------------------------------------------------------|
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |
|-------|------------------------------------------------------------------------------|

### RelatedPartyRefOrPartyRoleRef sub-resource fields

RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.

|                  |                                                                                                                                                                            |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| partyOrPartyRole | A PartyRefOrPartyRoleRef.                                                                                                                                                  |
| role             | A String [1]. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'. |
| @baseType        | A String. When sub-classing, this defines the super-class.                                                                                                                 |
| @schemaLocation  | A String. A URI to a JSON-Schema file that defines additional attributes and relationships.                                                                                |
| @type            | A String [1]. When sub-classing, this defines the sub-class Extensible name.                                                                                               |

### PartyRef sub-resource fields

A Party reference.

|                 |                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------|
| href            | A String. Hyperlink reference.                                                              |
| id              | A String [1]. Unique identifier.                                                            |
| name            | A String. Name of the referred entity.                                                      |
| @baseType       | A String. When sub-classing, this defines the super-class.                                  |
| @referredType   | A String. The actual type of the target instance when needed for disambiguation.            |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type           | A String [1].                                                                               |

### PartyRoleRef sub-resource fields

Party role reference. A party role represents the part played by a party in a given context.

|      |                                  |
|------|----------------------------------|
| href | A String. Hyperlink reference.   |
| id   | A String [1]. Unique identifier. |

|                 |                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------|
| name            | A String. Name of the referred entity.                                                      |
| partyId         | A String. The identifier of the engaged party that is linked to the PartyRole object.       |
| partyName       | A String. The name of the engaged party that is linked to the PartyRole object.             |
| @baseType       | A String. When sub-classing, this defines the super-class.                                  |
| @referredType   | A String. The actual type of the target instance when needed for disambiguation.            |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type           | A String [1].                                                                               |

### GeographicSiteFeature sub-resource fields

Geographic Site Feature captures various site information, ranging from survey data to safety guidelines and hazard information.

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| attachment            | An AttachmentRefOrValue [0..*]. Attachments relevant to this Feature, such as documentation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| featureCategory       | A String [0..*]. Collection of feature categories.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| featureCharacteristic | <p>A Characteristic [0..*]. This is a list of Characteristics for a particular feature.</p> <p>Characteristic can be instantiated as</p> <ul style="list-style-type: none"> <li>* BooleanArrayCharacteristic</li> <li>* BooleanCharacteristic</li> <li>* FloatCharacteristic</li> <li>* IntegerArrayCharacteristic</li> <li>* IntegerCharacteristic</li> <li>* NumberArrayCharacteristic</li> <li>* NumberCharacteristic</li> <li>* ObjectArrayCharacteristic</li> <li>* ObjectCharacteristic</li> <li>* StringArrayCharacteristic</li> <li>* StringCharacteristic</li> </ul> |
| featureRelationship   | A FeatureRelationship [0..*]. Collection of feature relationships.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| id                    | A String. Unique identifier.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                  |                                                                                             |
|------------------|---------------------------------------------------------------------------------------------|
| isBundle         | A Boolean. True if this is a feature group. Default is false.                               |
| isEnabled        | A Boolean. True if this feature is enabled. Default is true.                                |
| name             | A String [1]. The name for the feature.                                                     |
| note             | A Note [0..*]. Notes relevant to this Feature.                                              |
| policyConstraint | A PolicyRef [0..*]. Reference to manage a list of feature specification policy constraints. |
| relatedParty     | A RelatedPartyRefOrPartyRoleRef [1..*]. Party that may be relevant to this Feature.         |
| validFor         | A CalendarPeriod [0..*]. Validity period of this Feature.                                   |
| @baseType        | A String. When sub-classing, this defines the super-class.                                  |
| @schemaLocation  | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type            | A String [1]. When sub-classing, this defines the sub-class Extensible name.                |

### Attachment sub-resource fields

Complements the description of an element (for instance a product) through video, pictures...

|                |                                                                                                                                |
|----------------|--------------------------------------------------------------------------------------------------------------------------------|
| attachmentType | A String [1]. A business characterization of the purpose of the attachment, for example logo, instructionManual, contractCopy. |
| content        | A Base64. The actual contents of the attachment object, if embedded, encoded as base64.                                        |
| description    | A String. A narrative text describing the content of the attachment.                                                           |
| href           | A String. Hyperlink reference.                                                                                                 |
| id             | A String. Unique identifier.                                                                                                   |
| contentType    | A String [1]. A technical characterization of the attachment content format using IETF Mime Types.                             |
| name           | A String. The name of the attachment.                                                                                          |
| size           | A Quantity. An amount in a given unit.                                                                                         |
| url            | A String. Uniform Resource Locator, is a web page address (a subset of URI).                                                   |

|                 |                                                                                                        |
|-----------------|--------------------------------------------------------------------------------------------------------|
| validFor        | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both. |
| @baseType       | A String. When sub-classing, this defines the super-class.                                             |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships.            |
| @type           | A String [1]. When sub-classing, this defines the sub-class Extensible name.                           |

### AttachmentRef sub-resource fields

Attachment reference. An attachment complements the description of an element (for instance a product) through video, pictures.

|                 |                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------|
| description     | A String. A narrative text describing the content of the attachment.                        |
| href            | A String. Hyperlink reference.                                                              |
| id              | A String [1]. Unique identifier.                                                            |
| name            | A String. Name of the referred entity.                                                      |
| url             | A String. Link to the attachment media/content.                                             |
| @baseType       | A String. When sub-classing, this defines the super-class.                                  |
| @referredType   | A String [1]. The actual type of the target instance when needed for disambiguation.        |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type           | A String [1]. When sub-classing, this defines the sub-class Extensible name.                |

### Characteristic sub-resource fields

Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic...

|                            |                                                                                  |
|----------------------------|----------------------------------------------------------------------------------|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships. |
|----------------------------|----------------------------------------------------------------------------------|

|                 |                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| id              | A String. Unique identifier of the characteristic.                                                                                                                                                                                                                                                                                                              |
| name            | A String [1]. Name of the characteristic.                                                                                                                                                                                                                                                                                                                       |
| valueType       | A String. Data type of the value of the characteristic.                                                                                                                                                                                                                                                                                                         |
| @baseType       | A String. When sub-classing, this defines the super-class.                                                                                                                                                                                                                                                                                                      |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships.                                                                                                                                                                                                                                                                     |
| @type           | A String [1]. When sub-classing, this defines the sub-class Extensible name.                                                                                                                                                                                                                                                                                    |
|                 | Characteristic can be instantiated as<br>* BooleanArrayCharacteristic<br>* BooleanCharacteristic<br>* FloatCharacteristic<br>* IntegerArrayCharacteristic<br>* IntegerCharacteristic<br>* NumberArrayCharacteristic<br>* NumberCharacteristic<br>* ObjectArrayCharacteristic<br>* ObjectCharacteristic<br>* StringArrayCharacteristic<br>* StringCharacteristic |
| value           | This property is present in subclasses                                                                                                                                                                                                                                                                                                                          |

### CharacteristicRelationship sub-resource fields

Another Characteristic that is related to the current Characteristic;.

|                  |                                                                                             |
|------------------|---------------------------------------------------------------------------------------------|
| id               | A String [1]. Unique identifier of the characteristic.                                      |
| relationshipType | A String [1]. The type of relationship.                                                     |
| @baseType        | A String. When sub-classing, this defines the super-class.                                  |
| @schemaLocation  | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type            | A String [1]. When sub-classing, this defines the sub-class Extensible name.                |

### BooleanArrayCharacteristic sub-resource fields

A characteristic which value is an array of Boolean(s).

|                            |                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships.            |
| id                         | A String. Unique identifier of the characteristic.                                          |
| name                       | A String [1]. Name of the characteristic.                                                   |
| value                      | A Boolean [1..*]. A characteristic which value is an array of Boolean(s).                   |
| valueType                  | A String. Data type of the value of the characteristic.                                     |
| @baseType                  | A String. When sub-classing, this defines the super-class.                                  |
| @schemaLocation            | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type                      | A String [1]. When sub-classing, this defines the sub-class Extensible name.                |

### BooleanCharacteristic sub-resource fields

A characteristic which value is a Boolean.

|                            |                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships.            |
| id                         | A String. Unique identifier of the characteristic.                                          |
| name                       | A String [1]. Name of the characteristic.                                                   |
| value                      | A Boolean [1]. Value of the characteristic.                                                 |
| valueType                  | A String. Data type of the value of the characteristic.                                     |
| @baseType                  | A String. When sub-classing, this defines the super-class.                                  |
| @schemaLocation            | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type                      | A String [1]. When sub-classing, this defines the sub-class Extensible name.                |

### FloatCharacteristic sub-resource fields

A characteristic which value is a Float.

|                            |                                                                                  |
|----------------------------|----------------------------------------------------------------------------------|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships. |
| id                         | A String. Unique identifier of the characteristic.                               |

|                 |                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------|
| name            | A String [1]. Name of the characteristic.                                                   |
| value           | A Float [1]. Value of the characteristic.                                                   |
| valueType       | A String. Data type of the value of the characteristic.                                     |
| @baseType       | A String. When sub-classing, this defines the super-class.                                  |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type           | A String [1]. When sub-classing, this defines the sub-class Extensible name.                |

### IntegerArrayCharacteristic sub-resource fields

A characteristic which value is an array of Integer(s).

|                            |                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships.            |
| id                         | A String. Unique identifier of the characteristic.                                          |
| name                       | A String [1]. Name of the characteristic.                                                   |
| value                      | An Integer [1..*]. A characteristic which value is an array of Integer(s).                  |
| valueType                  | A String. Data type of the value of the characteristic.                                     |
| @baseType                  | A String. When sub-classing, this defines the super-class.                                  |
| @schemaLocation            | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type                      | A String [1]. When sub-classing, this defines the sub-class Extensible name.                |

### IntegerCharacteristic sub-resource fields

A characteristic which value is a Integer.

|                            |                                                                                  |
|----------------------------|----------------------------------------------------------------------------------|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships. |
| id                         | A String. Unique identifier of the characteristic.                               |
| name                       | A String [1]. Name of the characteristic.                                        |
| value                      | An Integer [1]. Value of the characteristic.                                     |
| valueType                  | A String. Data type of the value of the characteristic.                          |

|                 |                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------|
| @baseType       | A String. When sub-classing, this defines the super-class.                                  |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type           | A String [1]. When sub-classing, this defines the sub-class Extensible name.                |

### NumberArrayCharacteristic sub-resource fields

A characteristic which value is an array of Number(s).

|                            |                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships.            |
| id                         | A String. Unique identifier of the characteristic.                                          |
| name                       | A String [1]. Name of the characteristic.                                                   |
| value                      | A Number [1..*]. A characteristic which value is an array of Number(s).                     |
| valueType                  | A String. Data type of the value of the characteristic.                                     |
| @baseType                  | A String. When sub-classing, this defines the super-class.                                  |
| @schemaLocation            | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type                      | A String [1]. When sub-classing, this defines the sub-class Extensible name.                |

### NumberCharacteristic sub-resource fields

A characteristic which value is a Number.

|                            |                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships.            |
| id                         | A String. Unique identifier of the characteristic.                                          |
| name                       | A String [1]. Name of the characteristic.                                                   |
| value                      | A Number [1]. Value of the characteristic.                                                  |
| valueType                  | A String. Data type of the value of the characteristic.                                     |
| @baseType                  | A String. When sub-classing, this defines the super-class.                                  |
| @schemaLocation            | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |

|       |                                                                              |
|-------|------------------------------------------------------------------------------|
| @type | A String [1]. When sub-classing, this defines the sub-class Extensible name. |
|-------|------------------------------------------------------------------------------|

### ObjectArrayCharacteristic sub-resource fields

A characteristic which value is an array of Object(s).

|                            |                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships.            |
| id                         | A String. Unique identifier of the characteristic.                                          |
| name                       | A String [1]. Name of the characteristic.                                                   |
| value                      | An object [1..*]. Collection of characteristic values.                                      |
| valueType                  | A String. Data type of the value of the characteristic.                                     |
| @baseType                  | A String. When sub-classing, this defines the super-class.                                  |
| @schemaLocation            | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type                      | A String [1]. When sub-classing, this defines the sub-class Extensible name.                |

### ObjectCharacteristic sub-resource fields

A characteristic which value is a Object.

|                            |                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships.            |
| id                         | A String. Unique identifier of the characteristic.                                          |
| name                       | A String [1]. Name of the characteristic.                                                   |
| value                      | An object [1]. Value of the characteristic.                                                 |
| valueType                  | A String. Data type of the value of the characteristic.                                     |
| @baseType                  | A String. When sub-classing, this defines the super-class.                                  |
| @schemaLocation            | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type                      | A String [1]. When sub-classing, this defines the sub-class Extensible name.                |

### StringArrayCharacteristic sub-resource fields

A characteristic which value is an array of String(s).

|                            |                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships.            |
| id                         | A String. Unique identifier of the characteristic.                                          |
| name                       | A String [1]. Name of the characteristic.                                                   |
| value                      | A String [1..*]. Collection of string characteristics.                                      |
| valueType                  | A String. Data type of the value of the characteristic.                                     |
| @baseType                  | A String. When sub-classing, this defines the super-class.                                  |
| @schemaLocation            | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type                      | A String [1]. When sub-classing, this defines the sub-class Extensible name.                |

### StringCharacteristic sub-resource fields

A characteristic which value is a String.

|                            |                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------|
| characteristicRelationship | A CharacteristicRelationship [0..*]. Collection of characteristic relationships.            |
| id                         | A String. Unique identifier of the characteristic.                                          |
| name                       | A String [1]. Name of the characteristic.                                                   |
| value                      | A String [1]. Value of the characteristic.                                                  |
| valueType                  | A String. Data type of the value of the characteristic.                                     |
| @baseType                  | A String. When sub-classing, this defines the super-class.                                  |
| @schemaLocation            | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type                      | A String [1]. When sub-classing, this defines the sub-class Extensible name.                |

### FeatureRelationship sub-resource fields

Configuration feature.

|      |                                  |
|------|----------------------------------|
| href | A String. Hyperlink reference.   |
| id   | A String [1]. Unique identifier. |

|                  |                                                                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| name             | A String. This is the name of the target feature.                                                                                                 |
| relationshipType | A String [1]. This is the type of the feature relationship.<br>ENUMERATED with values:<br>* excluded<br>* includes<br>* may include<br>* requires |
| validFor         | A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.                                            |
| @baseType        | A String. When sub-classing, this defines the super-class.                                                                                        |
| @referredType    | A String. The actual type of the target instance when needed for disambiguation.                                                                  |
| @schemaLocation  | A String. A URI to a JSON-Schema file that defines additional attributes and relationships.                                                       |
| @type            | A String [1]. When sub-classing, this defines the sub-class Extensible name.                                                                      |

### PolicyRef sub-resource fields

Reference to managed Policy object.

|                 |                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------|
| href            | A String. Hyperlink reference.                                                              |
| id              | A String [1]. Unique identifier.                                                            |
| name            | A String. Name of the referred entity.                                                      |
| version         | A String.                                                                                   |
| @baseType       | A String. When sub-classing, this defines the super-class.                                  |
| @referredType   | A String. The actual type of the target instance when needed for disambiguation.            |
| @schemaLocation | A String. A URI to a JSON-Schema file that defines additional attributes and relationships. |
| @type           | A String [1]. When sub-classing, this defines the sub-class Extensible name.                |

### Feature sub-resource fields

Configuration feature.

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| attachment            | An AttachmentRefOrValue [0..*]. Attachments relevant to this Feature, such as documentation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| featureCategory       | A String [0..*]. Collection of feature categories.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| featureCharacteristic | <p>A Characteristic [0..*]. This is a list of Characteristics for a particular feature.</p> <p>Characteristic can be instantiated as</p> <ul style="list-style-type: none"> <li>* BooleanArrayCharacteristic</li> <li>* BooleanCharacteristic</li> <li>* FloatCharacteristic</li> <li>* IntegerArrayCharacteristic</li> <li>* IntegerCharacteristic</li> <li>* NumberArrayCharacteristic</li> <li>* NumberCharacteristic</li> <li>* ObjectArrayCharacteristic</li> <li>* ObjectCharacteristic</li> <li>* StringArrayCharacteristic</li> <li>* StringCharacteristic</li> </ul> |
| featureRelationship   | A FeatureRelationship [0..*]. Collection of feature relationships.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| id                    | A String. Unique identifier.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| isBundle              | A Boolean. True if this is a feature group. Default is false.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| isEnabled             | A Boolean. True if this feature is enabled. Default is true.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| name                  | A String [1]. The name for the feature.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| note                  | A Note [0..*]. Notes relevant to this Feature.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| policyConstraint      | A PolicyRef [0..*]. Reference to manage a list of feature specification policy constraints.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| relatedParty          | A RelatedPartyRefOrPartyRoleRef [1..*]. Party that may be relevant to this Feature.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| validFor              | A CalendarPeriod [0..*]. Validity period of this Feature.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| @baseType             | A String. When sub-classing, this defines the super-class.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| @schemaLocation       | A String. A URI to a JSON-Schema file that defines additional attributes and relationships.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| @type                 | A String [1]. When sub-classing, this defines the sub-class Extensible name.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|  |                                                           |
|--|-----------------------------------------------------------|
|  | Feature can be instantiated as<br>* GeographicSiteFeature |
|--|-----------------------------------------------------------|

## Json representation sample(s)

Geographic site resource example.

```
{
  "id": "ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b",
  "href": "https://mycsp.com/tmf-api/geographicSiteManagement/v5/geographicSite/ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b",
  "name": "BS-9283",
  "@type": "GeographicSite",
  "externalIdentifier": [
    {
      "id": "f76bc43c-53d5-4222-8ac3-f1a3e5b7d3ab",
      "@type": "ExternalIdentifier",
      "owner": "Address Management System",
      "externalIdentifierType": "Address"
    }
  ],
  "creationDate": "2024-03-15T07:44:40.056Z",
  "description": "This is the site for the base station BS-9283",
  "status": "active",
  "siteCategory": "BaseStation",
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Owner",
      "partyOrPartyRole": {
        "id": "da7f6b47-cd08-45a8-926b-d5824a5980e3",
        "href": "https://mycsp.com/tmf-api/partyManagement/v5/organization/da7f6b47-cd08-45a8-926b-d5824a5980e3",
        "name": "TelcoEngineering",
        "@type": "PartyRef",
        "@referredType": "Organization"
      }
    }
  ],
  "calendar": [
    {
      "@type": "CalendarPeriod",
      "day": "weekdays",
      "timeZone": "+00:00",
      "hourPeriod": [
        {
          "startHour": "09:00",
          "endHour": "18:00"
        }
      ],
      "status": "open"
    }
  ],
  "relatedAddress": [
    {
      "role": "service-address",

```

```

    "@type": "RelatedGeographicAddressRef",
    "address": {
      "id": "fcdb7157-3df9-4c49-93a6-8a3a248d9a27",
      "href": "https://mycsp.com/tmf-
api/geographicAddressManagement/v5/geographicAddress/fcdb7157-3df9-4c49-93a6-
8a3a248d9a27",
      "@type": "GeographicAddressRef",
      "@referredType": "GeographicAddress"
    }
  },
  ],
  "siteFeature": [
    {
      "id": "f9f06997-5a8c-4079-9c84-afdd154c3fcf",
      "name": "Out Of Hours Access",
      "@type": "GeographicSiteFeature",
      "featureCategory": [
        "OutOfHoursAccess"
      ],
      "isEnabled": true,
      "featureCharacteristic": [
        {
          "id": "4b69fcbb-6d89-4de3-8f90-fd12a567ddfc",
          "@type": "StringCharacteristic",
          "name": "Description",
          "value": "For out-of-hours weekday access please report to the reception to
gain access to the site."
        }
      ],
    },
    {
      "validFor": [
        {
          "@type": "CalendarPeriod",
          "day": "weekdays",
          "timeZone": "+00:00",
          "hourPeriod": [
            {
              "startHour": "18:00",
              "endHour": "09:00"
            }
          ],
        },
      ],
      "status": "Open"
    }
  ],
},
{
  "id": "d31a8f92-b8e7-4c82-8042-997ff5b64e30",
  "name": "Survey Information",
  "@type": "GeographicSiteFeature",
  "featureCategory": [
    "GeographicSiteFeature"
  ],
  "featureCharacteristic": [
    {
      "id": "4b69fcbb-6d89-4de3-8f90-fd12a567ddfc",
      "@type": "StringCharacteristic",
      "name": "Site Survey Reports",
      "value": "Information on underground and overhead utilities (e.g., gas lines,
electrical cables) to avoid interference during installation."
    }
  ],
},
],

```

```

    "attachment": [
      {
        "id": "bc4d58a3-549c-4082-98de-d1dc9b1066a4",
        "href": "https://mycsp.com/tmf-api/documentManagement/v5/attachment/bc4d58a3-549c-4082-98de-d1dc9b1066a4",
        "@referredType": "Attachment",
        "@type": "AttachmentRef"
      }
    ],
    "note": [
      {
        "id": "8921d8e6-6570-4260-bba8-e66cbba4b49b",
        "@type": "Note",
        "author": "Mr. N. Bene",
        "date": "2024-03-16T08:00:00Z",
        "text": "Site Survey received from TechSurvey&Co"
      }
    ],
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "SurveyPartner",
        "partyOrPartyRole": {
          "id": "1b3cc700-36cf-404c-85d5-c9f1c3c8fd99",
          "href": "https://mycsp.com/tmf-api/partyManagement/v5/organization/1b3cc700-36cf-404c-85d5-c9f1c3c8fd99",
          "name": "TechSurvey&Co",
          "@type": "PartyRef",
          "@referredType": "Organization"
        }
      }
    ]
  }
}

```

## Notification Resource Models

4 notifications are defined for this API.

Notifications related to GeographicSite:

- Create Event
- Delete Event
- Attribute Value Change Event
- State Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This

notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

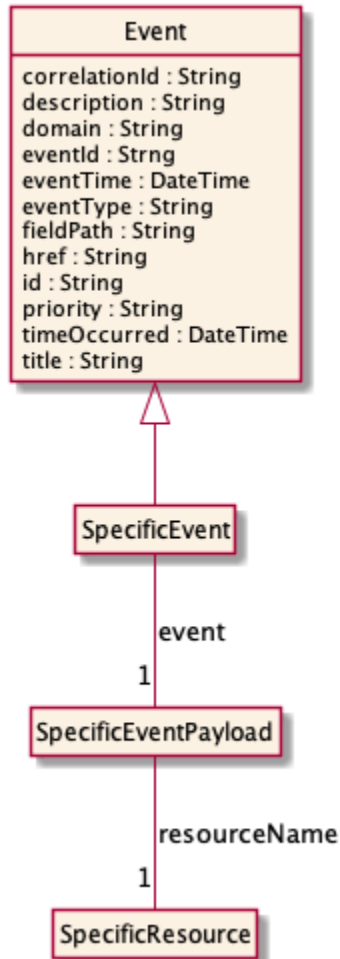


Figure 5 Notification Pattern

## GeographicSite

### Create Event

Message example for GeographicSiteCreateEvent event

```

Content-Type: application/json
{
  "correlationId": "82392652-42ce-4284-9a47-16d0fb06d269",
  "description": "GeographicSiteCreateEvent illustration",
  "domain": "Resource",
  "eventId": "43265751-ceca-4133-9338-31bfd4edbf75",
  "eventTime": "2024-08-25T12:25:22.380Z",
  "eventType": "GeographicSiteCreateEvent",
  "priority": "3",

```

```

"timeOccurred": "2022-08-25T12:25:12.746Z",
"title": "GeographicSiteCreateEvent",
"event": {
  "geographicSite": {
    "id": "ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b",
    "href": "https://mycsp.com/tmf-
api/geographicSiteManagement/v5/geographicSite/ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b",
    "@type": "GeographicSite",
    "externalIdentifier": [
      {
        "id": "f76bc43c-53d5-4222-8ac3-f1a3e5b7d3ab",
        "@type": "ExternalIdentifier",
        "owner": "Address Management System",
        "externalIdentifierType": "Address"
      }
    ],
    "creationDate": "2024-03-15T07:44:40.056Z",
    "description": "This is the site for the base station BS-9283",
    "status": "active",
    "siteCategory": "BaseStation",
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "Owner",
        "partyOrPartyRole": {
          "id": "da7f6b47-cd08-45a8-926b-d5824a5980e3",
          "href": "https://mycsp.com/tmf-api/partyManagement/v5/individual/da7f6b47-
cd08-45a8-926b-d5824a5980e3",
          "name": "TelcoEngineering",
          "@type": "PartyRef",
          "@referredType": "Organization"
        }
      }
    ],
    "calendar": [
      {
        "@type": "CalendarPeriod",
        "day": "weekdays",
        "timeZone": "+00:00",
        "hourPeriod": [
          {
            "endHour": "18:00",
            "startHour": "09:00"
          }
        ],
        "status": "open"
      }
    ],
    "relatedAddress": [
      {
        "role": "service-address",
        "address": {
          "id": "fcdb7157-3df9-4c49-93a6-8a3a248d9a27",
          "href": "https://mycsp.com/tmf-
api/geographicAddressManagement/v5/geographicAddress/fcdb7157-3df9-4c49-93a6-
8a3a248d9a27",
          "@type": "GeographicAddressRef",
          "@referredType": "GeographicAddress"
        },
        "@type": "RelatedGeographicAddressRef"
      }
    ]
  }
}

```

```

    }
  ],
  "siteFeature": [
    {
      "id": "f9f06997-5a8c-4079-9c84-afdd154c3fcf",
      "name": "Out Of Hours Access",
      "@type": "GeographicSiteFeature",
      "featureCategory": [
        "OutOfHoursAccess"
      ],
      "isEnabled": true,
      "featureCharacteristic": [
        {
          "id": "4b69fcbb-6d89-4de3-8f90-fd12a567ddfc",
          "@type": "StringCharacteristic",
          "name": "Out of Hours Access",
          "value": "For out-of-hours weekday access please report to the reception
to gain access to the site."
        }
      ],
      "validFor": [
        {
          "@type": "CalendarPeriod",
          "day": "weekdays",
          "timeZone": "+00:00",
          "hourPeriod": [
            {
              "startHour": "18:00",
              "endHour": "09:00"
            }
          ],
          "status": "open"
        }
      ]
    },
    {
      "id": "d31a8f92-b8e7-4c82-8042-997ff5b64e30",
      "name": "Survey Information",
      "@type": "GeographicSiteFeature",
      "featureCategory": [
        "SiteFeature"
      ],
      "featureCharacteristic": [
        {
          "id": "4b69fcbb-6d89-4de3-8f90-fd12a567ddfc",
          "@type": "StringCharacteristic",
          "name": "Site Survey Reports",
          "value": "Information on underground and overhead utilities (e.g., gas
lines, electrical cables) to avoid interference during installation."
        }
      ],
      "attachment": [
        {
          "id": "bc4d58a3-549c-4082-98de-d1dc9b1066a4",
          "href": "https://mycsp.com/tmf-
api/documentManagement/v5/attachment/bc4d58a3-549c-4082-98de-d1dc9b1066a4",
          "@referredType": "Attachment",
          "@type": "AttachmentRef"
        }
      ]
    }
  ],

```

```

    "note": [
      {
        "id": "8921d8e6-6570-4260-bba8-e66cbba4b49b",
        "@type": "Note",
        "author": "Mr. N. Bene",
        "date": "2024-03-16T08:00:00Z",
        "text": "Site Survey received from TechSurvey&Co"
      }
    ],
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "SurveyPartner",
        "partyOrPartyRole": {
          "id": "1b3cc700-36cf-404c-85d5-c9f1c3c8fd99",
          "href": "https://mycsp.com/tmf-api/partyManagement/v5/organization/1b3cc700-36cf-404c-85d5-c9f1c3c8fd99",
          "name": "TechSurvey&Co",
          "@type": "PartyRef",
          "@referredType": "Organization"
        }
      }
    ]
  }
},
"reportingSystem": {
  "id": "1afd23e9-5cfc-45b7-93a2-3de523b34971",
  "name": "APP-746",
  "@type": "ReportingResource",
  "@referredType": "LogicalResource"
},
"source": {
  "id": "769d82a4-c0c8-4d0d-8e76-c8b38de8b06d",
  "name": "APP-623",
  "@type": "ReportingResource",
  "@referredType": "LogicalResource"
},
"@baseType": "Event",
"@type": "GeographicSiteCreateEvent"
}

```

## Delete Event

Message example for GeographicSiteDeleteEvent event

```

Content-Type: application/json
{
  "correlationId": "92751b27-3994-45db-9893-b4b6a525bae8",
  "description": "GeographicSiteDeleteEvent illustration",
  "domain": "Commercial",
  "eventId": "3682a85c-1594-4153-a29e-fe4f94dfbb10",
  "eventTime": "2024-03-18T07:23:06Z",
  "eventType": "GeographicSiteDeleteEvent",
  "priority": "2",

```

```

"timeOccurred": "2024-03-18T07:23:06Z",
"title": "GeographicSiteDeleteEvent",
"event": {
  "geographicSite": {
    "id": "ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b",
    "href": "https://mycsp.com/tmf-
api/geographicSiteManagement/v5/geographicSite/ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b",
    "@type": "GeographicSite"
  }
},
"reportingSystem": {
  "id": "2759b03b-5d5c-42be-af24-015de991badf",
  "name": "APP-746",
  "@type": "ReportingResource",
  "@referredType": "LogicalResource"
},
"source": {
  "id": "db017150-c573-465a-8c39-47c2f730fe37",
  "name": "APP-623",
  "@type": "ReportingResource",
  "@referredType": "LogicalResource"
},
"@baseType": "Event",
"@type": "GeographicSiteDeleteEvent"
}

```

## Attribute Value Change Event

Message example for GeographicSiteAttributeValueChangeEvent event

```

Content-Type: application/json
{
  "correlationId": "e757af54-9a28-4f35-b2e7-071a6736c62d",
  "description": "GeographicSiteAttributeValueChangeEvent illustration",
  "domain": "Resource",
  "eventId": "64a81a78-9f37-42f3-a59d-a59b4699a7ee",
  "eventTime": "2024-08-25T12:25:22.389Z",
  "eventType": "GeographicSiteAttributeValueChangeEvent",
  "priority": "4",
  "timeOccurred": "2022-08-25T12:25:12.838Z",
  "title": "GeographicSiteAttributeValueChangeEvent",
  "event": {
    "geographicSite": {
      "id": "0738f85e-c8b0-4ed2-ac3e-b47687ade274",
      "href": "https://mycsp.com/tmf-
api/geographicSiteManagement/v5/geographicSite/0738f85e-c8b0-4ed2-ac3e-b47687ade274",
      "@type": "GeographicSite",
      "externalIdentifier": [
        {
          "id": "01abc593-82f5-4609-bd6e-31e58078e690",
          "@type": "ExternalIdentifier",
          "owner": "Address Management System",
          "externalIdentifierType": "Address"
        }
      ]
    }
  }
}

```

```

    },
    "reportingSystem": {
      "id": "cd8d6280-1abc-4542-8663-263fbe2c0eed",
      "name": "APP-746",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "source": {
      "id": "1235910b-2ed0-4d38-a5c2-ee35cb7cc542",
      "name": "APP-623",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "GeographicSiteAttributeValueChangeEvent"
  }
}

```

## State Change Event

Message example for GeographicSiteStateChangeEvent event

```

Content-Type: application/json
{
  "correlationId": "8dd2490d-8730-4110-a1a8-34bf9c1971a3",
  "description": "GeographicSiteStateChangeEvent illustration",
  "domain": "Resource",
  "eventId": "1d75bf59-3ec0-4d26-af4e-b77a13633771",
  "eventTime": "2024-08-25T12:25:22.399Z",
  "eventType": "GeographicSiteStateChangeEvent",
  "priority": "5",
  "timeOccurred": "2022-08-25T12:25:21.823Z",
  "title": "GeographicSiteStateChangeEvent",
  "event": {
    "geographicSite": {
      "id": "0738f85e-c8b0-4ed2-ac3e-b47687ade274",
      "href": "https://mycsp.com/tmf-api/geographicSiteManagement/v5/geographicSite/0738f85e-c8b0-4ed2-ac3e-b47687ade274",
      "@type": "GeographicSite",
      "status": "active"
    }
  },
  "reportingSystem": {
    "id": "d20571a2-d2a5-4676-99ab-025636682ed8",
    "name": "APP-746",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "50a04a29-f37b-439c-933d-329e40b3beef",
    "name": "APP-623",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "GeographicSiteStateChangeEvent"
}

```

}

## API OPERATIONS

Remember the following Uniform Contract:

| Operation on Entities          | Uniform API Operation | Description                                                    |
|--------------------------------|-----------------------|----------------------------------------------------------------|
| Query Entities                 | GET Resource          | GET must be used to retrieve a representation of a resource.   |
| Create Entity                  | POST Resource         | POST must be used to create a new resource                     |
| Partial Update of an Entity    | PATCH Resource        | PATCH must be used to partially update a resource              |
| Remove an Entity               | DELETE Resource       | DELETE must be used to remove a resource                       |
| Execute an Action on an Entity | POST on TASK Resource | POST must be used to execute Task Resources                    |
| Other Request Methods          | POST on TASK Resource | GET and POST must not be used to tunnel other request methods. |

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

### Operations on GeographicSite

#### Retrieves a GeographicSite by ID

**GET** /geographicSite/{id}?fields=...

#### Description

This operation retrieves a GeographicSite entity. Attribute selection is enabled for all first level attributes.

#### Usage samples

Retrieving a geographic site with an address defined by reference (id).

#### Request

```
GET /geographicSite/ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b
Content-Type: application/json
```

## Response

```
200
{
  "@type": "GeographicSite",
  "href": "https://mycsp.com/tmf-api/geographicSiteManagement/v5/geographicSite/ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b",
  "id": "ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b",
  "name": "BS-9283",
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "owner": "Address Management System",
      "externalIdentifierType": "Address",
      "id": "f76bc43c-53d5-4222-8ac3-f1a3e5b7d3ab"
    }
  ],
  "creationDate": "2024-03-15T07:44:40.056Z",
  "description": "This is the site for the base station BS-9283",
  "status": "active",
  "siteCategory": "BaseStation",
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Owner",
      "partyOrPartyRole": {
        "href": "https://mycsp.com/tmf-api/partyManagement/v5/individual/da7f6b47-cd08-45a8-926b-d5824a5980e3",
        "id": "da7f6b47-cd08-45a8-926b-d5824a5980e3",
        "name": "TelcoEngineering",
        "@type": "PartyRef",
        "@referredType": "Organization"
      }
    }
  ],
  "calendar": [
    {
      "@type": "CalendarPeriod",
      "day": "weekdays",
      "timeZone": "+00:00",
      "hourPeriod": [
        {
          "startHour": "09:00",
          "endHour": "18:00"
        }
      ],
      "status": "open"
    }
  ],
  "relatedAddress": [
    {
      "role": "service-address",
      "address": {
```

```

        "id": "fcdb7157-3df9-4c49-93a6-8a3a248d9a27",
        "href": "https://mycsp.com/tmf-
api/geographicAddressManagement/v5/geographicAddress/fcdb7157-3df9-4c49-93a6-
8a3a248d9a27",
        "@type": "GeographicAddressRef",
        "@referredType": "GeographicAddress"
    },
    "@type": "RelatedGeographicAddressRef"
}
],
"siteFeature": [
{
    "id": "f9f06997-5a8c-4079-9c84-afdd154c3fcf",
    "name": "Out Of Hours Access",
    "@type": "GeographicSiteFeature",
    "featureCategory": [
        "OutOfHoursAccess"
    ],
    "isEnabled": true,
    "featureCharacteristic": [
        {
            "@type": "StringCharacteristic",
            "id": "4b69fcbb-6d89-4de3-8f90-fd12a567ddfc",
            "name": "Description",
            "value": "For out-of-hours weekday access please report to the reception to
gain access to the site."
        }
    ],
    "validFor": [
        {
            "@type": "CalendarPeriod",
            "day": "weekdays",
            "timeZone": "+00:00",
            "hourPeriod": [
                {
                    "startHour": "18:00",
                    "endHour": "09:00"
                }
            ],
            "status": "open"
        }
    ]
},
{
    "id": "d31a8f92-b8e7-4c82-8042-997ff5b64e30",
    "name": "Survey Information",
    "@type": "GeographicSiteFeature",
    "featureCategory": [
        "GeographicSiteFeature"
    ],
    "featureCharacteristic": [
        {
            "@type": "StringCharacteristic",
            "id": "4b69fcbb-6d89-4de3-8f90-fd12a567ddfc",
            "name": "Site Survey Reports",
            "value": "Information on underground and overhead utilities (e.g., gas lines,
electrical cables) to avoid interference during installation."
        }
    ],
    "attachment": [

```

```

    {
      "href": "https://mycsp.com/tmf-api/documentManagement/v5/attachment/bc4d58a3-549c-4082-98de-d1dc9b1066a4",
      "id": "bc4d58a3-549c-4082-98de-d1dc9b1066a4",
      "@referredType": "Attachment",
      "@type": "AttachmentRef"
    }
  ],
  "note": [
    {
      "@type": "Note",
      "id": "8921d8e6-6570-4260-bba8-e66cbba4b49b",
      "author": "Mr. N. Bene",
      "date": "2024-03-16T08:00:00Z",
      "text": "Site Survey received from TechSurvey&Co"
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "SurveyPartner",
      "partyOrPartyRole": {
        "href": "https://mycsp.com/tmf-api/partyManagement/v5/organization/1b3cc700-36cf-404c-85d5-c9f1c3c8fd99",
        "id": "1b3cc700-36cf-404c-85d5-c9f1c3c8fd99",
        "name": "TechSurvey&Co",
        "@type": "PartyRef",
        "@referredType": "Organization"
      }
    }
  ]
}

```

## List or find GeographicSite objects

**GET /geographicSite?fields=...&{filtering}**

### Description

This operation list GeographicSite entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

### Usage samples

Retrieving a list of geographic sites with a siteCategory (BaseStation) and status (active).

### Request

```
GET /geographicSite?filter=siteCategory=BaseStation,status=active&fields=id,href,name
```

```
Content-Type: application/json
```

## Response

```
200
[
  {
    "@type": "GeographicSite",
    "href": "https://mycsp.com/tmf-api/geographicSiteManagement/v5/geographicSite/ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b",
    "id": "ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b",
    "name": "BS-9283"
  },
  {
    "@type": "GeographicSite",
    "href": "https://mycsp.com/tmf-api/geographicSiteManagement/v5/geographicSite/0738f85e-c8b0-4ed2-ac3e-b47687ade274",
    "id": "0738f85e-c8b0-4ed2-ac3e-b47687ade274",
    "name": "BS-2397"
  }
]
```

## Creates a GeographicSite

**POST /geographicSite?fields=...&{filtering}**

### Description

This operation creates a GeographicSite entity.

### Mandatory Attributes

| Mandatory Attributes | Rule |
|----------------------|------|
| name                 |      |
| @type                |      |

### Usage samples

Creation of a new geographic site with POST operation

### Request

```
POST /geographicSite
Content-Type: application/json
{
  "@type": "GeographicSite",
  "name": "BS-9283",
  "externalIdentifier": [
```

```

    {
      "@type": "ExternalIdentifier",
      "owner": "Address Management System",
      "externalIdentifierType": "Address",
      "id": "f76bc43c-53d5-4222-8ac3-f1a3e5b7d3ab"
    }
  ],
  "description": "This is the site for the base station BS-9283",
  "status": "active",
  "siteCategory": "BaseStation",
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Owner",
      "partyOrPartyRole": {
        "href": "https://mycsp.com/tmf-api/partyManagement/v5/individual/da7f6b47-cd08-45a8-926b-d5824a5980e3",
        "id": "da7f6b47-cd08-45a8-926b-d5824a5980e3",
        "name": "TelcoEngineering",
        "@type": "PartyRef",
        "@referredType": "Organization"
      }
    }
  ],
  "calendar": [
    {
      "@type": "CalendarPeriod",
      "day": "weekdays",
      "timeZone": "+00:00",
      "hourPeriod": [
        {
          "startHour": "09:00",
          "endHour": "18:00"
        }
      ],
      "status": "open"
    }
  ],
  "relatedAddress": [
    {
      "role": "service-address",
      "address": {
        "id": "fcdb7157-3df9-4c49-93a6-8a3a248d9a27",
        "href": "https://mycsp.com/tmf-api/geographicAddressManagement/v5/geographicAddress/fcdb7157-3df9-4c49-93a6-8a3a248d9a27",
        "@type": "GeographicAddressRef",
        "@referredType": "GeographicAddress"
      },
      "@type": "RelatedGeographicAddressRef"
    }
  ],
  "siteFeature": [
    {
      "id": "f9f06997-5a8c-4079-9c84-afdd154c3fcf",
      "name": "Out Of Hours Access",
      "@type": "GeographicSiteFeature",
      "featureCategory": [
        "OutOfHoursAccess"
      ]
    }
  ],

```

```

    "isEnabled": true,
    "featureCharacteristic": [
      {
        "@type": "StringCharacteristic",
        "id": "4b69fcbb-6d89-4de3-8f90-fd12a567ddfc",
        "name": "Out of Hours Access",
        "value": "For out-of-hours weekday access please report to the reception to
gain access to the site."
      }
    ],
    "validFor": [
      {
        "@type": "CalendarPeriod",
        "day": "weekdays",
        "timeZone": "+00:00",
        "hourPeriod": [
          {
            "startHour": "18:00",
            "endHour": "09:00"
          }
        ],
        "status": "open"
      }
    ]
  },
  {
    "id": "d31a8f92-b8e7-4c82-8042-997ff5b64e30",
    "name": "Survey Information",
    "@type": "GeographicSiteFeature",
    "featureCategory": [
      "GeographicSiteFeature"
    ],
    "featureCharacteristic": [
      {
        "@type": "StringCharacteristic",
        "id": "4b69fcbb-6d89-4de3-8f90-fd12a567ddfc",
        "name": "Site Survey Reports",
        "value": "Information on underground and overhead utilities (e.g., gas lines,
electrical cables) to avoid interference during installation."
      }
    ],
    "attachment": [
      {
        "href": "https://mycsp.com/tmf-api/documentManagement/v5/attachment/bc4d58a3-
549c-4082-98de-d1dc9b1066a4",
        "id": "bc4d58a3-549c-4082-98de-d1dc9b1066a4",
        "@referredType": "Attachment",
        "@type": "AttachmentRef"
      }
    ],
    "note": [
      {
        "@type": "Note",
        "id": "8921d8e6-6570-4260-bba8-e66cbba4b49b",
        "author": "Mr. N. Bene",
        "date": "2024-03-16T08:00:00Z",
        "text": "Site Survey received from TechSurvey&Co"
      }
    ],
    "relatedParty": [

```

```

    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "SurveyPartner",
      "partyOrPartyRole": {
        "href": "https://mycsp.com/tmf-
api/partyManagement/v5/organization/1b3cc700-36cf-404c-85d5-c9f1c3c8fd99",
        "id": "1b3cc700-36cf-404c-85d5-c9f1c3c8fd99",
        "name": "TechSurvey&Co",
        "@type": "PartyRef",
        "@referredType": "Organization"
      }
    }
  ]
}

```

## Response

```

201
{
  "@type": "GeographicSite",
  "href": "https://mycsp.com/tmf-
api/geographicSiteManagement/v5/geographicSite/f9ee9a78-63e1-4e52-aef9-8102afe3da95",
  "id": "f9ee9a78-63e1-4e52-aef9-8102afe3da95",
  "name": "BS-9283",
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "owner": "Address Management System",
      "externalIdentifierType": "Address",
      "id": "f76bc43c-53d5-4222-8ac3-f1a3e5b7d3ab"
    }
  ],
  "creationDate": "2024-03-15T07:44:40.056Z",
  "description": "This is the site for the base station BS-9283",
  "status": "active",
  "siteCategory": "BaseStation",
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "role": "Owner",
      "partyOrPartyRole": {
        "href": "https://mycsp.com/tmf-api/partyManagement/v5/individual/da7f6b47-cd08-
45a8-926b-d5824a5980e3",
        "id": "da7f6b47-cd08-45a8-926b-d5824a5980e3",
        "name": "TelcoEngineering",
        "@type": "PartyRef",
        "@referredType": "Organization"
      }
    }
  ],
  "calendar": [
    {
      "@type": "CalendarPeriod",
      "day": "weekdays",
      "timeZone": "+00:00",

```

```

    "hourPeriod": [
      {
        "startHour": "09:00",
        "endHour": "18:00"
      }
    ],
    "status": "open"
  }
],
"relatedAddress": [
  {
    "role": "service-address",
    "address": {
      "id": "fcdb7157-3df9-4c49-93a6-8a3a248d9a27",
      "href": "https://mycsp.com/tmf-
api/geographicAddressManagement/v5/geographicAddress/fcdb7157-3df9-4c49-93a6-
8a3a248d9a27",
      "@type": "GeographicAddressRef",
      "@referredType": "GeographicAddress"
    },
    "@type": "RelatedGeographicAddressRef"
  }
],
"siteFeature": [
  {
    "id": "f9f06997-5a8c-4079-9c84-afdd154c3fcf",
    "name": "Out Of Hours Access",
    "@type": "GeographicSiteFeature",
    "featureCategory": [
      "OutOfHoursAccess"
    ],
    "isEnabled": true,
    "featureCharacteristic": [
      {
        "@type": "StringCharacteristic",
        "id": "4b69fcbb-6d89-4de3-8f90-fd12a567ddfc",
        "name": "Out of Hours Access",
        "value": "For out-of-hours weekday access please report to the reception to
gain access to the site."
      }
    ],
    "validFor": [
      {
        "@type": "CalendarPeriod",
        "day": "weekdays",
        "timeZone": "+00:00",
        "hourPeriod": [
          {
            "startHour": "18:00",
            "endHour": "09:00"
          }
        ],
        "status": "open"
      }
    ]
  },
  {
    "id": "d31a8f92-b8e7-4c82-8042-997ff5b64e30",
    "name": "Survey Information",
    "@type": "GeographicSiteFeature",

```

```

    "featureCategory": [
      "GeographicSiteFeature"
    ],
    "featureCharacteristic": [
      {
        "@type": "StringCharacteristic",
        "id": "4b69fcbb-6d89-4de3-8f90-fd12a567ddfc",
        "name": "Site Survey Reports",
        "value": "Information on underground and overhead utilities (e.g., gas lines,
electrical cables) to avoid interference during installation."
      }
    ],
    "attachment": [
      {
        "href": "https://mycsp.com/tmf-api/documentManagement/v5/attachment/bc4d58a3-
549c-4082-98de-d1dc9b1066a4",
        "id": "bc4d58a3-549c-4082-98de-d1dc9b1066a4",
        "@referredType": "Attachment",
        "@type": "AttachmentRef"
      }
    ],
    "note": [
      {
        "@type": "Note",
        "id": "8921d8e6-6570-4260-bba8-e66cbba4b49b",
        "author": "Mr. N. Bene",
        "date": "2024-03-16T08:00:00Z",
        "text": "Site Survey received from TechSurvey&Co"
      }
    ],
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "SurveyPartner",
        "partyOrPartyRole": {
          "href": "https://mycsp.com/tmf-
api/partyManagement/v5/organization/1b3cc700-36cf-404c-85d5-c9f1c3c8fd99",
          "id": "1b3cc700-36cf-404c-85d5-c9f1c3c8fd99",
          "name": "TechSurvey&Co",
          "@type": "PartyRef",
          "@referredType": "Organization"
        }
      }
    ]
  }
}

```

## Updates partially a GeographicSite

**PATCH /geographicSite/{id}?fields=...**

### Description

This operation allows partial updates of a GeographicSite entity. Support of json/merge (<https://tools.ietf.org/html/rfc7396>) is mandatory, support of json/patch

(<http://tools.ietf.org/html/rfc5789>) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

### Patchable and Non Patchable Attributes

| Non Patchable Attributes | Rule                         |
|--------------------------|------------------------------|
| creationDate             |                              |
| href                     |                              |
| id                       |                              |
| lastUpdate               |                              |
| @baseType                | @baseType is immutable       |
| @schemaLocation          | @schemaLocation is immutable |
| @type                    | @type is immutable           |

| Patchable Attributes       | Rule |
|----------------------------|------|
| calendar                   |      |
| capacity                   |      |
| code                       |      |
| contactMedium              |      |
| description                |      |
| externalIdentifier         |      |
| geographicSiteRelationship |      |
| name                       |      |
| note                       |      |
| relatedAddress             |      |
| relatedLocation            |      |
| relatedParty               |      |
| siteCategory               |      |
| siteFeature                |      |
| status                     |      |

## Usage samples

Here is an example of a request for updating geographic site ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b – Change site category and description (JSON Patch; update applied only if all `test` operations succeed) – the full resource representation is sent in PATCH response

### Request

```
PATCH /geographicSite/ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b
Content-Type: application/json-patch+json
[
  {
    "op": "test",
    "path": "/siteCategory",
    "value": "franchise-store"
  },
  {
    "op": "replace",
    "path": "/siteCategory",
    "value": "owned-store"
  },
  {
    "op": "test",
    "path": "/description",
    "value": "This is the site for the base station BS-9283"
  },
  {
    "op": "replace",
    "path": "/description",
    "value": "This is the site for the base station BS-9999"
  }
]
```

### Response

```
200
{
  "@type": "GeographicSite",
  "href": "https://mycsp.com/tmf-api/geographicSiteManagement/v5/geographicSite/ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b",
  "id": "ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b",
  "name": "BS-9999",
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "owner": "Address Management System",
      "externalIdentifierType": "Address",
      "id": "f76bc43c-53d5-4222-8ac3-f1a3e5b7d3ab"
    }
  ],
  "creationDate": "2024-03-15T07:44:40.056Z",
  "description": "This is the site for the base station BS-9999",
  "status": "active",
  "siteCategory": "owned-store",
}
```

```

"relatedParty": [
  {
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "role": "Owner",
    "partyOrPartyRole": {
      "href": "https://mycsp.com/tmf-api/partyManagement/v5/individual/da7f6b47-cd08-45a8-926b-d5824a5980e3",
      "id": "da7f6b47-cd08-45a8-926b-d5824a5980e3",
      "name": "TelcoEngineering",
      "@type": "PartyRef",
      "@referredType": "Organization"
    }
  }
],
"calendar": [
  {
    "@type": "CalendarPeriod",
    "day": "weekdays",
    "timeZone": "+00:00",
    "hourPeriod": [
      {
        "startHour": "09:00",
        "endHour": "18:00"
      }
    ],
    "status": "open"
  }
],
"relatedAddress": [
  {
    "role": "service-address",
    "address": {
      "id": "fcdb7157-3df9-4c49-93a6-8a3a248d9a27",
      "href": "https://mycsp.com/tmf-api/geographicAddressManagement/v5/geographicAddress/fcdb7157-3df9-4c49-93a6-8a3a248d9a27",
      "@type": "GeographicAddressRef",
      "@referredType": "GeographicAddress"
    },
    "@type": "RelatedGeographicAddressRef"
  }
],
"siteFeature": [
  {
    "id": "f9f06997-5a8c-4079-9c84-afdd154c3fcf",
    "name": "Out Of Hours Access",
    "@type": "GeographicSiteFeature",
    "featureCategory": [
      "OutOfHoursAccess"
    ],
    "isEnabled": true,
    "featureCharacteristic": [
      {
        "@type": "StringCharacteristic",
        "id": "4b69fcbb-6d89-4de3-8f90-fd12a567ddfc",
        "name": "Description",
        "value": "For out-of-hours weekday access please report to the reception to gain access to the site."
      }
    ]
  }
],

```

```

    "validFor": [
      {
        "@type": "CalendarPeriod",
        "day": "weekdays",
        "timeZone": "+00:00",
        "hourPeriod": [
          {
            "startHour": "18:00",
            "endHour": "09:00"
          }
        ],
        "status": "open"
      }
    ],
    {
      "id": "d31a8f92-b8e7-4c82-8042-997ff5b64e30",
      "name": "Survey Information",
      "@type": "GeographicSiteFeature",
      "featureCategory": [
        "GeographicSiteFeature"
      ],
      "featureCharacteristic": [
        {
          "@type": "StringCharacteristic",
          "id": "4b69fcbb-6d89-4de3-8f90-fd12a567ddfc",
          "name": "Site Survey Reports",
          "value": "Information on underground and overhead utilities (e.g., gas lines,
electrical cables) to avoid interference during installation."
        }
      ],
      "attachment": [
        {
          "href": "https://mycsp.com/tmf-api/documentManagement/v5/attachment/bc4d58a3-
549c-4082-98de-d1dc9b1066a4",
          "id": "bc4d58a3-549c-4082-98de-d1dc9b1066a4",
          "@referredType": "Attachment",
          "@type": "AttachmentRef"
        }
      ],
      "note": [
        {
          "@type": "Note",
          "id": "8921d8e6-6570-4260-bba8-e66cbba4b49b",
          "author": "Mr. N. Bene",
          "date": "2024-03-16T08:00:00Z",
          "text": "Site Survey received from TechSurvey&Co"
        }
      ],
      "relatedParty": [
        {
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "role": "SurveyPartner",
          "partyOrPartyRole": {
            "href": "https://mycsp.com/tmf-
api/partyManagement/v5/organization/1b3cc700-36cf-404c-85d5-c9f1c3c8fd99",
            "id": "1b3cc700-36cf-404c-85d5-c9f1c3c8fd99",
            "name": "TechSurvey&Co",
            "@type": "PartyRef",
            "@referredType": "Organization"
          }
        }
      ]
    }
  ]
}

```

```

    }
  }
]
}

```

Here is an example of a request for updating geographic site ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b – Change start and end hour, and site category – the full resource representation is sent in PATCH response

### Request

```

PATCH /geographicSite/ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b
Content-Type: application/merge-patch+json
{
  "siteCategory": "Warehouse",
  "calendar": [
    {
      "@type": "CalendarPeriod",
      "day": "weekdays",
      "timeZone": "+00:00",
      "hourPeriod": [
        {
          "startHour": "07:00",
          "endHour": "22:00"
        }
      ],
      "status": "open"
    }
  ],
  "@type": "GeographicSite"
}

```

### Response

```

200
{
  "@type": "GeographicSite",
  "href": "https://mycsp.com/tmf-api/geographicSiteManagement/v5/geographicSite/ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b",
  "id": "ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b",
  "name": "WSH-2837",
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "owner": "Address Management System",
      "externalIdentifierType": "Address",
      "id": "f76bc43c-53d5-4222-8ac3-f1a3e5b7d3ab"
    }
  ],
  "creationDate": "2024-03-15T07:44:40.056Z",
  "description": "This is the site for the warehouse WSH-2837",
}

```

```

"status": "active",
"siteCategory": "Warehouse",
"relatedParty": [
  {
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "role": "Owner",
    "partyOrPartyRole": {
      "href": "https://mycsp.com/tmf-api/partyManagement/v5/individual/da7f6b47-cd08-45a8-926b-d5824a5980e3",
      "id": "da7f6b47-cd08-45a8-926b-d5824a5980e3",
      "name": "TelcoEngineering",
      "@type": "PartyRef",
      "@referredType": "Organization"
    }
  }
],
"calendar": [
  {
    "@type": "CalendarPeriod",
    "day": "weekdays",
    "timeZone": "+00:00",
    "hourPeriod": [
      {
        "startHour": "07:00",
        "endHour": "22:00"
      }
    ],
    "status": "open"
  }
],
"relatedAddress": [
  {
    "role": "service-address",
    "address": {
      "id": "fcdb7157-3df9-4c49-93a6-8a3a248d9a27",
      "href": "https://mycsp.com/tmf-api/geographicAddressManagement/v5/geographicAddress/fcdb7157-3df9-4c49-93a6-8a3a248d9a27",
      "@type": "GeographicAddressRef",
      "@referredType": "GeographicAddress"
    },
    "@type": "RelatedGeographicAddressRef"
  }
],
"siteFeature": [
  {
    "id": "f9f06997-5a8c-4079-9c84-afdd154c3fcf",
    "name": "Out Of Hours Access",
    "@type": "GeographicSiteFeature",
    "featureCategory": [
      "OutOfHoursAccess"
    ],
    "isEnabled": true,
    "featureCharacteristic": [
      {
        "@type": "StringCharacteristic",
        "id": "4b69fcbb-6d89-4de3-8f90-fd12a567ddfc",
        "name": "Description",
        "value": "For out-of-hours weekday access please report to the reception to gain access to the site."
      }
    ]
  }
]

```

```

    }
  ],
  "validFor": [
    {
      "@type": "CalendarPeriod",
      "day": "weekdays",
      "timeZone": "+00:00",
      "hourPeriod": [
        {
          "startHour": "18:00",
          "endHour": "09:00"
        }
      ],
      "status": "open"
    }
  ],
  {
    "id": "d31a8f92-b8e7-4c82-8042-997ff5b64e30",
    "name": "Survey Information",
    "@type": "GeographicSiteFeature",
    "featureCategory": [
      "GeographicSiteFeature"
    ],
    "featureCharacteristic": [
      {
        "@type": "StringCharacteristic",
        "id": "4b69fcbb-6d89-4de3-8f90-fd12a567ddfc",
        "name": "Site Survey Reports",
        "value": "Information on underground and overhead utilities (e.g., gas lines,
electrical cables) to avoid interference during installation."
      }
    ],
    "attachment": [
      {
        "href": "https://mycsp.com/tmf-api/documentManagement/v5/attachment/bc4d58a3-
549c-4082-98de-d1dc9b1066a4",
        "id": "bc4d58a3-549c-4082-98de-d1dc9b1066a4",
        "@referredType": "Attachment",
        "@type": "AttachmentRef"
      }
    ],
    "note": [
      {
        "@type": "Note",
        "id": "8921d8e6-6570-4260-bba8-e66cbba4b49b",
        "author": "Mr. N. Bene",
        "date": "2024-03-16T08:00:00Z",
        "text": "Site Survey received from TechSurvey&Co"
      }
    ],
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "SurveyPartner",
        "partyOrPartyRole": {
          "href": "https://mycsp.com/tmf-
api/partyManagement/v5/organization/1b3cc700-36cf-404c-85d5-c9f1c3c8fd99",
          "id": "1b3cc700-36cf-404c-85d5-c9f1c3c8fd99",
          "name": "TechSurvey&Co",

```

```

        "@type": "PartyRef",
        "@referredType": "Organization"
      }
    ]
  }
}

```

Here is an example of a request for updating geographic site ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b – Change start and end hour, and site category – the full resource representation is sent in PATCH response

### Request

```

PATCH /geographicSite/ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b
Content-Type: application/merge-patch+json
{
  "siteCategory": "Warehouse",
  "calendar": [
    {
      "@type": "CalendarPeriod",
      "day": "weekdays",
      "timeZone": "+00:00",
      "hourPeriod": [
        {
          "endHour": "22:00",
          "startHour": "07:00"
        }
      ],
      "status": "open"
    }
  ],
  "@type": "GeographicSite"
}

```

### Response

```

200
{
  "@type": "GeographicSite",
  "href": "https://mycsp.com/tmf-api/geographicSiteManagement/v5/geographicSite/ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b",
  "id": "ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b",
  "name": "WSH-2837",
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "owner": "Address Management System",
      "externalIdentifierType": "Address",
      "id": "f76bc43c-53d5-4222-8ac3-f1a3e5b7d3ab"
    }
  ],
}

```

```

"creationDate": "2024-03-15T07:44:40.056Z",
"description": "This is the site for the warehouse WSH-2837",
"status": "active",
"siteCategory": "Warehouse",
"relatedParty": [
  {
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "role": "Owner",
    "partyOrPartyRole": {
      "href": "https://mycsp.com/tmf-api/partyManagement/v5/individual/da7f6b47-cd08-45a8-926b-d5824a5980e3",
      "id": "da7f6b47-cd08-45a8-926b-d5824a5980e3",
      "name": "TelcoEngineering",
      "@type": "PartyRef",
      "@referredType": "Organization"
    }
  }
],
"calendar": [
  {
    "@type": "CalendarPeriod",
    "day": "weekdays",
    "timeZone": "+00:00",
    "hourPeriod": [
      {
        "endHour": "22:00",
        "startHour": "07:00"
      }
    ]
  }
],
"status": "open"
},
"relatedAddress": [
  {
    "role": "service-address",
    "address": {
      "id": "fcdb7157-3df9-4c49-93a6-8a3a248d9a27",
      "href": "https://mycsp.com/tmf-api/geographicAddressManagement/v5/geographicAddress/fcdb7157-3df9-4c49-93a6-8a3a248d9a27",
      "@type": "GeographicAddressRef",
      "@referredType": "GeographicAddress"
    },
    "@type": "RelatedGeographicAddressRef"
  }
],
"siteFeature": [
  {
    "id": "f9f06997-5a8c-4079-9c84-afdd154c3fcf",
    "name": "Out Of Hours Access",
    "@type": "GeographicSiteFeature",
    "featureCategory": [
      "OutOfHoursAccess"
    ],
    "isEnabled": true,
    "featureCharacteristic": [
      {
        "@type": "StringCharacteristic",
        "id": "4b69fcbb-6d89-4de3-8f90-fd12a567ddfc",
        "name": "Description",

```

```

        "value": "For out-of-hours weekday access please report to the reception to
gain access to the site."
    }
],
"validFor": [
    {
        "@type": "CalendarPeriod",
        "day": "weekdays",
        "timeZone": "+00:00",
        "hourPeriod": [
            {
                "startHour": "18:00",
                "endHour": "09:00"
            }
        ],
        "status": "open"
    }
]
},
{
    "id": "d31a8f92-b8e7-4c82-8042-997ff5b64e30",
    "name": "Survey Information",
    "@type": "GeographicSiteFeature",
    "featureCategory": [
        "GeographicSiteFeature"
    ],
    "featureCharacteristic": [
        {
            "@type": "StringCharacteristic",
            "id": "4b69fcbb-6d89-4de3-8f90-fd12a567ddfc",
            "name": "Site Survey Reports",
            "value": "Information on underground and overhead utilities (e.g., gas lines,
electrical cables) to avoid interference during installation."
        }
    ],
    "attachment": [
        {
            "href": "https://mycsp.com/tmf-api/documentManagement/v5/attachment/bc4d58a3-
549c-4082-98de-d1dc9b1066a4",
            "id": "bc4d58a3-549c-4082-98de-d1dc9b1066a4",
            "@referredType": "Attachment",
            "@type": "AttachmentRef"
        }
    ],
    "note": [
        {
            "@type": "Note",
            "id": "8921d8e6-6570-4260-bba8-e66cbba4b49b",
            "author": "Mr. N. Bene",
            "date": "2024-03-16T08:00:00Z",
            "text": "Site Survey received from TechSurvey&Co"
        }
    ],
    "relatedParty": [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "SurveyPartner",
            "partyOrPartyRole": {
                "href": "https://mycsp.com/tmf-
api/partyManagement/v5/organization/1b3cc700-36cf-404c-85d5-c9f1c3c8fd99",

```

```

        "id": "1b3cc700-36cf-404c-85d5-c9f1c3c8fd99",
        "name": "TechSurvey&Co",
        "@type": "PartyRef",
        "@referredType": "Organization"
      }
    ]
  }
}

```

Here is an example of a request for updating geographic site ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b – Change a site feature name (JSON Patch Query; update applied only if all `test` operations succeed) – the full resource representation is sent in PATCH response

### Request

```

PATCH /geographicSite/ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b
Content-Type: application/json-patch-query+json
[
  {
    "op": "test",
    "path": "$.siteFeature[?(@.id=='f9f06997-5a8c-4079-9c84-afdd154c3fcf')].name",
    "value": "Out Of Hours Access"
  },
  {
    "op": "replace",
    "path": "$.siteFeature[?(@.id=='f9f06997-5a8c-4079-9c84-afdd154c3fcf')].name",
    "value": "Out Of Hours Access Updated"
  }
]

```

### Response

```

200
{
  "@type": "GeographicSite",
  "href": "https://mycsp.com/tmf-api/geographicSiteManagement/v5/geographicSite/ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b",
  "id": "ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b",
  "name": "BS-9999",
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "owner": "Address Management System",
      "externalIdentifierType": "Address",
      "id": "f76bc43c-53d5-4222-8ac3-f1a3e5b7d3ab"
    }
  ],
  "creationDate": "2024-03-15T07:44:40.056Z",
  "description": "This is the site for the base station BS-9999",
  "status": "active",
  "siteCategory": "FiberExchange",

```

```

"relatedParty": [
  {
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "role": "Owner",
    "partyOrPartyRole": {
      "href": "https://mycsp.com/tmf-api/partyManagement/v5/individual/da7f6b47-cd08-45a8-926b-d5824a5980e3",
      "id": "da7f6b47-cd08-45a8-926b-d5824a5980e3",
      "name": "TelcoEngineering",
      "@type": "PartyRef",
      "@referredType": "Organization"
    }
  }
],
"calendar": [
  {
    "@type": "CalendarPeriod",
    "day": "weekdays",
    "timeZone": "+00:00",
    "hourPeriod": [
      {
        "startHour": "09:00",
        "endHour": "18:00"
      }
    ],
    "status": "open"
  }
],
"relatedAddress": [
  {
    "role": "service-address",
    "address": {
      "id": "fcdb7157-3df9-4c49-93a6-8a3a248d9a27",
      "href": "https://mycsp.com/tmf-api/geographicAddressManagement/v5/geographicAddress/fcdb7157-3df9-4c49-93a6-8a3a248d9a27",
      "@type": "GeographicAddressRef",
      "@referredType": "GeographicAddress"
    },
    "@type": "RelatedGeographicAddressRef"
  }
],
"siteFeature": [
  {
    "id": "f9f06997-5a8c-4079-9c84-afdd154c3fcf",
    "name": "Out Of Hours Access Updated",
    "@type": "GeographicSiteFeature",
    "featureCategory": [
      "OutOfHoursAccess"
    ],
    "isEnabled": true,
    "featureCharacteristic": [
      {
        "@type": "StringCharacteristic",
        "id": "4b69fcbb-6d89-4de3-8f90-fd12a567ddfc",
        "name": "Description",
        "value": "For out-of-hours weekday access please report to the reception to gain access to the site."
      }
    ]
  }
],

```

```

    "validFor": [
      {
        "@type": "CalendarPeriod",
        "day": "weekdays",
        "timeZone": "+00:00",
        "hourPeriod": [
          {
            "startHour": "18:00",
            "endHour": "09:00"
          }
        ],
        "status": "open"
      }
    ],
    {
      "id": "d31a8f92-b8e7-4c82-8042-997ff5b64e30",
      "name": "Survey Information",
      "@type": "GeographicSiteFeature",
      "featureCategory": [
        "GeographicSiteFeature"
      ],
      "featureCharacteristic": [
        {
          "@type": "StringCharacteristic",
          "id": "4b69fcbb-6d89-4de3-8f90-fd12a567ddfc",
          "name": "Site Survey Reports",
          "value": "Information on underground and overhead utilities (e.g., gas lines,
electrical cables) to avoid interference during installation."
        }
      ],
      "attachment": [
        {
          "href": "https://mycsp.com/tmf-api/documentManagement/v5/attachment/bc4d58a3-
549c-4082-98de-d1dc9b1066a4",
          "id": "bc4d58a3-549c-4082-98de-d1dc9b1066a4",
          "@referredType": "Attachment",
          "@type": "AttachmentRef"
        }
      ],
      "note": [
        {
          "@type": "Note",
          "id": "8921d8e6-6570-4260-bba8-e66cbba4b49b",
          "author": "Mr. N. Bene",
          "date": "2024-03-16T08:00:00Z",
          "text": "Site Survey received from TechSurvey&Co"
        }
      ],
      "relatedParty": [
        {
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "role": "SurveyPartner",
          "partyOrPartyRole": {
            "href": "https://mycsp.com/tmf-
api/partyManagement/v5/organization/1b3cc700-36cf-404c-85d5-c9f1c3c8fd99",
            "id": "1b3cc700-36cf-404c-85d5-c9f1c3c8fd99",
            "name": "TechSurvey&Co",
            "@type": "PartyRef",
            "@referredType": "Organization"
          }
        }
      ]
    }
  ]
}

```

```
}  
}  
]  
}  
]  
}
```

## Deletes a GeographicSite

**DELETE /geographicSite/{id}?{filtering}**

### Description

This operation deletes a GeographicSite entity.

### Usage samples

Here's an example of a request for delete a specific GeographicSite.

### Request

```
DELETE /geographicSite/ae4b915f-16f5-4f05-a3c7-8c6a99c7d24b  
Content-Type: application/json
```

### Response

```
204
```

## API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

### Register listener

#### POST /hub

##### Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

##### Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

##### Usage Samples

Here's an example of a request for registering a listener.

##### Request

```
POST /api/hub

Accept: application/json

{
  "callback": "http://in.listener.com"
}
```

##### Response

```
201

Content-Type: application/json

Location: /api/hub/42
```

```
{  
  "id": "42",  
  "callback": "http://in.listener.com",  
  "query": ""  
}
```

## Unregister listener

**DELETE /hub/{id}**

### Description

Clears the communication endpoint address that was set by creating the Hub..

### Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

### Usage Samples

Here's an example of a request for un-registering a listener.

### Request

```
DELETE /api/hub/42  
  
Accept: application/json
```

### Response

```
204
```

## Publish Event to listener

**POST /client/listener**

### Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

## Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

## Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

## Request

```
POST /client/listener

Accept: application/json

{
  "event": {
    EVENT BODY
  },
  "eventType": "EVENT_TYPE"
}
```

## Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

# Acknowledgements

## Release History

| Release Number | Date       | Release led by:                                                           | Description                 |
|----------------|------------|---------------------------------------------------------------------------|-----------------------------|
| Release 17.5   | 22-05-2017 | P. Gauthier / M.Belaunde/<br>L.Robert                                     |                             |
| Release 19.0   |            | P. Gauthier / L.Robert                                                    | Version 4.0 of the API REST |
| Version 5.0    | 01-05-2026 | Knut Johannessen, Florin<br>Tene, Rochana<br>Macharoen, Stephen<br>Harrop | Version 5.0 of the REST API |

## Contributors to Document

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